Online Supplement

What Prices Reveal About Hidden Trader Populations, and What Identifying Them Costs

The supplement contains the proofs (Section S1), further results and robustness checks, the extended literature review, the hypothetical dollar illustration, a finite-horizon decision example, the spectral blind spot, the measurement of reduced-form feedback and the reproducibility map. Section, table, figure and equation numbers prefixed with S refer to this supplement.

S1 Proofs

Lemma 2.1. (i) For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance P solves $P = P\sigma_\nu^2/(P+\sigma_\nu^2) + \sigma_\zeta^2$, i.e. $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$, whose positive root is $P = \big(\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}\big)/2$. The gain is $\eta = P/(P+\sigma_\nu^2) \in (0,1)$, which simplifies to (7), and the update $\hat{v}_{t+1} = \hat{v}_t + \eta(r_t - \hat{v}_t)$ has the stated form. (ii) $\nabla_\theta \frac{1}{2}H(x_t-\theta)^2 = -H(x_t-\theta)$, so $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t - \theta_t) = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H$ and $q = \alpha H/\eta$.

Lemma 2.3. Omit zero-weight cohorts. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}\big(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots\big), \qquad u = \big(\sqrt{a}, \sqrt{b_1\eta_1}, \ldots\big)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to κu and $J'e_x = \sqrt{a}\,u$, so $h_j = (\kappa/a)e_x^\top (J')^{j+1} e_x$. The spectral decomposition of J' gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0,1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is completely monotone. Since J is nonnegative with row sums at most one, every price row of J^t has ℓ_1, hence ℓ_2, norm at most one, attained at $t = 0$.

Theorem 3.1. With zero initial states, $y_{i,t} = \sum_{s\le t} \eta_i r_i^{t-s} x_s$, so $\sum_i q_i y_{i,t} = \sum_{s\le t} k_{t-s} x_s$ and every response is determined by (a, κ, k), hence by M. Conversely, equal responses for all probes imply equal H, hence equal G, and partial fractions of G identify the distinct poles with nonzero residues w_i, i.e. M. After cancellation, $H = \kappa z D(z)/[(z-a)D(z) - \kappa N(z)]$ with $D = \prod(z - r_i)$ and $N(z) = \sum_i w_i z \prod_{j\neq i}(z - r_j)$, as in (4); at a retained r_i the denominator equals $-\kappa N(r_i) = -\kappa w_i r_i \prod_{j\neq i}(r_i - r_j) \neq 0$ (memory rates are positive because $\eta_i < 1$), so no root of D cancels. The remaining numerator factor z cancels if and only if the denominator vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for

$a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1 + |\mathrm{supp}(M) \cap (0,1)|$ for $a' \neq 0$ and $|\mathrm{supp}(M) \cap (0,1)|$ for $a' = 0$ (an atom at rate 0, a cohort with $\eta = 1$, contributes the constant $M(\{0\})$ to G and no pole, so it shifts a to $a' = a + \kappa M(\{0\})$ and the argument above applies with a') (numerically, the rank of the Hankel matrix of $(h_j)_{j\geq 1}$ is $K+1$ and K respectively for $K = 1, 2, 3$). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z-1+\bar{\eta})\big((z-1+\bar{\eta})^2 - \delta^2\big)}.$$

On $|z| = 1$ each factor has modulus at least ϵ. Uniform stability gives $M_H = \sup|H_\theta| < \infty$, and the identity $H_\delta - H_0 = z^{-1} H_\delta H_0 (G_\delta - G_0)$, whose factor z^{-1} has modulus one on the circle, gives an ℓ_2 gain at most $C_{\mathcal{F}}\delta^2$. The mean difference of a power-constrained probe has squared norm at most $C_{\mathcal{F}}^2 \delta^4 PT$; the equal-noise KL is at most $C_{\mathcal{F}}^2 \delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (9).

Theorem 4.1. In either positive economy $J \geq 0$ and $\|J\|_\infty \leq 1 - \epsilon(1-S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1-S)$. From the display above, $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \leq K_* \delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \leq K_* \delta^2 \|v\|_1$, and Gaussian KL with Pinsker gives (10). For (ii), the mean difference is a lower-triangular convolution matrix times v; each column is a shifted copy of a prefix of d_T, so $\|\mu_\delta - \mu_0\|_2 \leq V \|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters h_2 with factor κ^2 while h_0, h_1 coincide; a pulse A separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the ℓ_2 gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t |\lambda x_t| \leq |\lambda| \kappa V$, attained by a single pulse.

Lemma 4.2. After the uncontrolled update, the pre-probe displacement at step j is $\sum_{i<j} h_{j-i} v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2} v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \leq \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

Theorem 4.3. Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \leq x_t \leq \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x_{j+1}^0 - \frac{\kappa}{2} \sum_{j<t} v_j^2 + c U_t$. Its

quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of h bounds every entry by $\kappa/2$. Hence $B - E_t \le \kappa U_t^2/2 + cU_t \le \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (14). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse A funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

Proposition 4.4. The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2\kappa^2 q\delta^2 > 0$.

Lemma 4.5. Marking residual inventory at P_* gives $E_\infty = B - \frac{1}{2}v^\top T_h v - c\sum|v_t| \ge 0$. For each mode $\lambda \le r$ the Toeplitz kernel has spectral density $(1-\lambda^2)/(1 - 2\lambda\cos\omega + \lambda^2) \ge (1-r)/(1+r)$, so $v^\top T_h v \ge m\|v\|_2^2$ and $\|v\|_2^2 \le 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2\delta^4\|v\|_2^2$, hence the KL bound; Pinsker gives (15). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory A; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu|I_t|P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \le 1$.

Proposition S9.1. Parts (i) and (ii) follow from (S2) with $\Delta h = \Theta(\delta)$ for a rate shift and $\Theta(\delta^2)$ for symmetric splitting, and from additivity of KL over independent pulse cycles. Part (iii) solves the session constraint for δ; substituting $d_T(\delta) \approx \delta^2 \tilde{d}_T$ gives the fourth-root approximation.

Theorems S9.2 and S9.3. For a flat round trip, telescoping the cash account gives $L_i(v) = \frac{1}{2}v^\top T_{h_i} v = \frac{1}{2\pi}\int g_i(\omega)|V(e^{i\omega})|^2 d\omega$, with $g_i > 0$ by Lemma 2.3. Parseval gives $\|(H_1 - H_0)v\|_2^2 = \frac{1}{2\pi}\int |\Delta\hat{H}|^2|V|^2 \le \sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, and the Neyman–Pearson requirement $\|(H_1 - H_0)v\|_2 \ge 2\sigma z_{1-\alpha}$ gives (S6). For adaptive policies, Doob's optional stopping theorem applied to the log-likelihood-ratio martingale and the data-processing inequality give $\mathbb{E}_0[\Lambda_\tau] \ge \mathrm{kl}(\alpha, 1-\beta)$, while $\mathbb{E}_0[\Lambda_\tau] = \mathbb{E}_0\sum_t(\Delta\mu_t)^2/(2\sigma^2) \le \sup(|\Delta\hat{H}|^2/g_0)\mathbb{E}_0[L_0]/(2\sigma^2)$, which gives (S8).

Part (iv). Let G_θ be the $T \times T$ lower-triangular Toeplitz section of H_θ/κ, so the observation covariance is $\Sigma_\theta = \sigma_w^2 G_\theta G_\theta^\top + \sigma_\nu^2 I$. A finite section of a bounded analytic symbol has norm at most its supremum on the circle, so $\|G_\theta\| \le \sup_\omega |H_\theta|/\kappa \le \bar{M}$ because $|H_\theta(e^{i\omega})| \le \sum_j h_j = \kappa/(1-S)$. The inverse of a lower-triangular Toeplitz section is the section of the reciprocal series; κ/H_θ is analytic on and outside the unit circle because H_θ is minimum phase (Theorem 5.7), and Lemma 2.3 gives $\mathrm{Re}\, H_\theta(e^{i\omega}) = \sum_\ell \omega_\ell(1 - \lambda_\ell\cos\omega)/(1 - 2\lambda_\ell\cos\omega + \lambda_\ell^2) \ge \sum_\ell \omega_\ell/(1+\lambda_\ell) \ge \kappa/2$, so $\|G_\theta^{-1}\| \le 2$ and $\Sigma_\theta \succeq \sigma_{\mathrm{eff}}^2 I$

(the smallest singular value of G_θ is at least 0.522 in 762 random positive economies and sections). The Gaussian divergence splits into a mean and a covariance part. The mean part is $\frac{1}{2}\Delta\mu^\top\Sigma_1^{-1}\Delta\mu \le \|(H_1-H_0)v\|_2^2/(2\sigma_{\text{eff}}^2)$. For the covariance part, the eigenvalues λ_k of $A=\Sigma_1^{-1/2}\Sigma_0\Sigma_1^{-1/2}$ are at least $c=\sigma_\nu^2/(\sigma_\nu^2+\sigma_w^2\bar{M}^2)$, and $\lambda-1-\log\lambda \le (\lambda-1)^2/\lambda$, so it is at most $\|A-I\|_F^2/(2c) \le \|\Sigma_0-\Sigma_1\|_F^2/(2c\,\sigma_{\text{eff}}^4)$. Writing $\Sigma_0-\Sigma_1=\sigma_w^2(\Delta G\,G_0^\top+G_1\Delta G^\top)$ and $\|\Delta G\|_F^2 \le T\|h^\delta-h^0\|_2^2/\kappa^2 \le TK_*^2\delta^4/\kappa^2$ (proof of Theorem 4.1) gives a covariance part of at most $C_wT\delta^4$. The data-processing inequality turns errors (α,β) into $D_{\text{KL}} \ge \text{kl}(\alpha,1-\beta)$, and part (ii) bounds $\|(H_1-H_0)v\|_2^2$ by $\sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, which gives (S7).

Theorem S9.4. Write a feasible flat probe as $v=Eu$ and expand $u=\sum_j c_je_j$ in the G_i-orthonormal eigenbasis, so $L_i(v)=\sum_j c_j^2$ and the separation constraint reads $\sum_j\lambda_jc_j^2 \ge c_{\text{req}}$. Write the top-two component as $(c_{L-2},c_{L-1})=r(\cos\theta,\sin\theta)$ with $r\ge 0$; replacing θ by $\theta+\pi$ flips d and s_θ and leaves (S9) unchanged, so $\theta\in[0,\pi)$ suffices. Put $L_0(\theta)=c_{\text{req}}/\lambda(\theta)$ and $D=L_i(v)-L_0(\theta)$. The constraint gives $r^2\lambda(\theta) \ge c_{\text{req}}-\sum_{j\le L-3}\lambda_jc_j^2$, hence

$$D=r^2+\sum_{j\le L-3}c_j^2-L_0(\theta) \;\ge\; \sum_{j\le L-3}\Big(1-\frac{\lambda_j}{\lambda(\theta)}\Big)c_j^2 \;\ge\; \Big(1-\frac{\lambda_{L-3}}{\lambda(\theta)}\Big)\sum_{j\le L-3}c_j^2 \;\ge 0,$$

and $r^2 \ge L_0(\theta)-\frac{\lambda_{L-3}}{\lambda(\theta)}\sum_{j\le L-3}c_j^2 \ge L_0(\theta)\big(1-\gamma(\theta)D\big)$. By Hölder's inequality with the sign vector s_θ, $c_{\text{fee}}\|Eu\|_1 \ge c_{\text{fee}}r\|Ed(\theta)\|_1+c_{\text{fee}}\sum_{j\le L-3}c_j\,s_\theta^\top Ee_j$, and Cauchy–Schwarz with weights $1-\lambda_j/\lambda(\theta)$ bounds the second term below by $-\sqrt{D\,W_\perp(\theta)}$. Using $r \ge \sqrt{L_0(1-\gamma D)_+} \ge \sqrt{L_0}\,(1-\gamma D)_+$ and the fact that the fee is nonnegative, with $x=\sqrt{D}$,

$$L_i(v)+c_{\text{fee}}\|v\|_1 \;\ge\; L_0(\theta)+x^2+\Big[F_{\text{dom}}(\theta)(1-\gamma(\theta)x^2)_+-\sqrt{W_\perp(\theta)}\,x\Big]_+ \;\ge\; L_0(\theta)+\Psi(\theta).$$

Minimizing over the unknown θ gives (S10), and $\Psi\ge 0$, $L_0(\theta)\ge c_{\text{req}}/\lambda_{L-1}$ give the last inequality. On the piece $\gamma x^2\le 1$ with $\gamma F_{\text{dom}}<1$ the bracket is a convex quadratic in x with minimum $F_{\text{dom}}-W_\perp/(4(1-\gamma F_{\text{dom}}))$, which gives the closed form.

Corollary S9.5. Since $\Delta h_t=O(\delta^2)$, the variance difference $\sigma_{1,t}^2-\sigma_{0,t}^2=\sigma^2\xi[(h_1+h_0)*v]_t[\Delta h*v]_t=O(\delta^2)$, and the variance term of the Gaussian KL, $\frac{1}{2}\sum(x-1-\log x)$ with $x=\sigma_{0,t}^2/\sigma_{1,t}^2$, is quadratic in that difference. Convex impact only increases dissipation.

Proposition S7.1. Standard Cramér–Rao for a Gaussian mean model with Jacobian J. For $K=2$ the columns of J for the two rates become collinear as $\Delta\eta\to 0$; expanding to third order, the smallest singular value of J is of order $(\Delta\eta)^3$, so $[F^{-1}]_{kk}=\Theta((\Delta\eta)^{-6})$.

Theorem 5.1. Expanding $M_m(\theta_1) - M_m(\theta_0)$ under (16) gives zero for $m \le 2$ and $4\tau p_- p_+ (\Delta\eta)^3$ for $m = 3$. The map from moments to impulse responses is analytic with nonsingular derivative at the common limit, so $\|\Delta h\|_2 = \Theta(\tau(\Delta\eta)^3)$. Le Cam's two-point lemma with N independent Gaussian experiments gives (18). For the permutation-invariant loss, the identity matching of θ_0 and θ_1 costs $2\tau + (p_+ - p_-)\Delta\eta = 2\tau(1 + \Delta\eta/q)$ per atom and the swapped matching costs $\max(2p_+, 2p_-)\Delta\eta = (1 + 2\tau/q)\Delta\eta$; when both are at least 2τ, any estimate within τ of one economy in d_∞ is farther than τ from the other by the triangle inequality, and the two-point argument is unchanged. For (iii), the upper bound is Theorem 5.4; the Prony–Hankel determinant identity follows from $g_m = w_1\eta_1 r_1^m + w_2\eta_2 r_2^m$.

Theorem 5.3. Let (s^A, ω^A) and (s^B, ω^B) be the K-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K - 2$, and their moments of degree $2K - 1$ differ by $g_K \neq 0$. With $r = r_c + \Delta\eta\, s$ and weights $W\omega$, the raw moments $\sum w_i r_i^m$ are polynomials in $\Delta\eta$ that agree below degree $2K - 1$, so the kernels differ by $O((\Delta\eta)^{2K-1})$ with nonzero leading coefficient; analyticity of the kernel-to-response map gives $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, and Le Cam's lemma gives the bound. For the W_1 version, the two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta\, s$ scaled to mass W, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

Lemma 5.5. Let $L = 2n - 2$ and let p^* be the best uniform approximation of $|x|$ on $[-1, 1]$ by polynomials of degree L, with error $E = E_L(|x|)$. By Chebyshev's equioscillation theorem (Rivlin, 1981) there are points $t_0 < \cdots < t_{L+1}$ at which $|x| - p^*(x)$ equals $\pm E$ with alternating signs. The divided-difference weights $w_i = 1/\prod_{j \neq i}(t_i - t_j)$ alternate in sign and satisfy $\sum_i w_i p(t_i) = 0$ for every polynomial p of degree at most L; multiplying by -1 if necessary, w_i has the sign of $|t_i| - p^*(t_i)$ for every i. Exactly n weights are positive and n negative, and the positive and negative parts have equal mass because constants are annihilated. Normalizing each part to unit mass and mapping $x \mapsto \rho(x + 1)/2$ gives n-atom probability measures π_A (positive part) and π_B (negative part) on $[0, \rho]$ with equal moments of orders $0, \ldots, L$. For the 1-Lipschitz function $f(r) = |r - \rho/2| = (\rho/2)|x|$,

$$\int f\, d(\pi_A - \pi_B) = \frac{\rho}{2} \int \big(|x| - p^*(x)\big)\, d(\pi_A - \pi_B) = \frac{\rho}{2}(E + E) = \rho E,$$

because p^* integrates to the same value under both measures, $|x| - p^* = E$ on the support of π_A and $-E$ on that of π_B. By Kantorovich–Rubinstein duality $W_1(\pi_A, \pi_B) \ge \rho E \ge c_* \rho / L$.

Theorem 5.6. Write $K(z) = \sum_m k_m z^m$, $D = 1 - az - \kappa z K$ and $H = \kappa/D$, which is the generating function of the recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ with $h_0 = \kappa$. Since $k_m \geq 0$ and $\sum_m k_m = \int (1-r)^{-1} d\mu \leq W/(1-\rho)$, the ℓ_1 norm of $az + \kappa z K$ is at most $s < 1$, so $\|H\|_1 \leq \kappa/(1-s)$ and $\|1/H\|_1 = \|D\|_1/\kappa \leq (1+s)/\kappa$.

(i) Let $\mu_A = W\pi_A$ and $\mu_B = W\pi_B$ with π_A, π_B from Lemma 5.5; both lie in $\mathcal{M}(\rho, W)$. With $\nu = \mu_A - \mu_B$, $\sum_m (k_m^A - k_m^B) z^m = \int (1-rz)^{-1} d\nu(r) = z^{2n-1} \int r^{2n-1} (1-rz)^{-1} d\nu(r)$, because the first $2n-1$ terms of the geometric series integrate to zero. On $|z| = 1$ the modulus is at most $2W\rho^{2n-1}/(1-\rho)$, so Parseval gives $\|k^A - k^B\|_2 \leq 2W\rho^{2n-1}/(1-\rho)$. Next, $H_A - H_B = z(K_A - K_B) H_A H_B$, so by Young's inequality $\|h^A - h^B\|_2 \leq \|H_A\|_1 \|H_B\|_1 \|k^A - k^B\|_2 \leq C_h \|k^A - k^B\|_2$. The Kullback–Leibler divergence between the N-sample Gaussian experiments is $N\|h^A - h^B\|_2^2/(2\sigma^2) \leq (N/\sigma^2)\, 2C_h^2 W^2 \rho^{2(2n-1)}/(1-\rho)^2$, which is at most one as soon as $2(2n-1)\log(1/\rho) \geq \Lambda_N - 2\log(1/\rho)$. Take n the smallest integer with this property and $n \geq 2$; then $2n - 2 < \Lambda_N/(2\log(1/\rho))$ when $\Lambda_N > 4\log(1/\rho)$. Since d is a norm on signed measures and $d(\mu_A, \mu_B) = W\, W_1(\pi_A, \pi_B) \geq W c_* \rho/(2n-2) > 2c_* \rho W \log(1/\rho)/\Lambda_N$, the Bretagnolle–Huber form of Le Cam's lemma (Tsybakov, 2009, Lemma 2.6 and Theorem 2.2) gives $\max_{j \in \{A,B\}} P_j(d(\hat{\mu}, \mu_j) \geq c_* \rho W \log(1/\rho)/\Lambda_N) \geq \frac{1}{4} e^{-1}$.

(ii) *Approximation.* Let f be 1-Lipschitz on $[0, \rho]$ with $f(0) = 0$, so $|f| \leq \rho$. The function $g(x) = f(\rho(x+1)/2)$ is $(\rho/2)$-Lipschitz on $[-1, 1]$, and Jackson's theorem (Rivlin, 1981) gives a polynomial p of degree L with $\sup|g - p| \leq (\pi/2)(\rho/2)/(L+1) \leq \rho$, hence $\sup|p| \leq 2\rho$. Its Chebyshev coefficients satisfy $|b_j| \leq 2 \sup|p| \leq 4\rho$. The polynomial $q(r) = p(2r/\rho - 1)$ approximates f with the same error, and its monomial coefficients satisfy $\sum_j |q_j| \leq \sum_{j \leq L} |b_j|\, \|T_j(2r/\rho - 1)\|_{\text{coef}} \leq 4\rho \sum_{j \leq L} B(\rho)^j \leq \rho B(\rho)^{L+1}$, where $\|\cdot\|_{\text{coef}}$ is the sum of absolute monomial coefficients, $\|T_j\|_{\text{coef}} \leq (1+\sqrt{2})^j$, substituting $y = \alpha r - 1$ multiplies the coefficient sum of y^i by at most $(1+\alpha)^i$, and $B(\rho) - 1 \geq 4$. On the event $\max_{j \leq L} |\hat{k}_j - k_j| \leq \delta$ the true μ attains objective at most δ, so the minimizer $\hat{\mu}$ satisfies $\max_{j \leq L} |\int r^j d\hat{\mu} - k_j| \leq 2\delta$ by the triangle inequality. *Moment comparison.* If $\max_{j \leq L} |\int r^j d\hat{\mu} - k_j| \leq 2\delta$, then $\hat{\mu}$ has mass at most $W + 2\delta$ and

$$\int f\, d(\hat{\mu} - \mu) = \int (f - q)\, d(\hat{\mu} - \mu) + \sum_{j \leq L} q_j \Big(\int r^j d\hat{\mu} - k_j \Big) \leq \frac{\pi\rho(W + \delta)}{2(L+1)} + 2\delta\rho B(\rho)^{L+1}.$$

Kernel recovery. From $\kappa z K = 1 - az - \kappa/H$, the kernel values $k_0, \ldots, k_L$ are the coefficients of $(1 - az - \kappa/H)/(\kappa z)$, which depend on $h_0, \ldots, h_{L+1}$ only. Work in the algebra of power series modulo z^{L+2} with the ℓ_1 norm, which is submultiplicative. If the averaged response is $\hat{H} = H + E$ and $\|1/H\|_1 \|E\|_1 \leq \frac{1}{2}$, then $1/\hat{H} = (1/H)(1 + E/H)^{-1}$ and $\|1/\hat{H} - 1/H\|_1 \leq 2\|1/H\|_1^2 \|E\|_1$, so $\max_j |\hat{k}_j - k_j| \leq \delta \equiv 2((1+s)/\kappa)^2 (L+2) \max_{t \leq L+1} |E_t|$. Each E_t is Gaussian with variance σ^2/N, so with probability at least $1 - \beta$, $\max_t |E_t| \leq$

$\sigma\sqrt{2\log(2(L+2)/\beta)/N}$. The true μ is feasible for the linear program with value at most δ, so the minimizer satisfies the moment condition above. *Choice of L.* With $L+1 = \lfloor \log(N/\sigma^2)/(4\log B(\rho))\rfloor$, $B(\rho)^{L+1} \le (N/\sigma^2)^{1/4}$, so $\delta B(\rho)^{L+1} = O((\sigma^2/N)^{1/4} L\sqrt{\log(L/\beta)})$ and $\delta/(L+1)$ are $o(1/\log(N/\sigma^2))$, while $\pi\rho W/(2(L+1)) \le 2\pi\rho W \log B(\rho)/(\log(N/\sigma^2) - 4\log B(\rho))$. For N/σ^2 larger than a threshold depending on $(\rho, W, a, \kappa, \beta)$ the side condition $\|1/H\|_1\|E\|_1 \le \frac{1}{2}$ holds on the event and the total is below $3\pi\rho W \log B(\rho)/\log(N/\sigma^2)$, uniformly over $\mathcal{M}(\rho, W)$. The final statement combines (i) and (ii).

Theorem 5.8. For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi}\int_{-\pi}^{\pi} \big(\varrho - 1 - \log\varrho\big)\, d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = z^{-1}H_1\Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(z^{-1}H_1\Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k \phi(z, \eta_k)$, with w_k the cohort weights (any factor of η_k is absorbed in ϕ), where ϕ is analytic in η on a neighbourhood of $[\eta_{\min}, \eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1 - \eta$ stays inside the disk). Expanding ϕ around $\bar\eta$, $\Delta G = \sum_{j\ge 0} \frac{\partial_\eta^j \phi(z,\bar\eta)}{j!}\Delta\tilde M_j$, where $\tilde M_j = \sum_k w_k(\eta_k - \bar\eta)^j$. For the pair, $\Delta\tilde M_0 = \Delta\tilde M_1 = 0$ and $\Delta\tilde M_2 = q\delta^2$; for the moment-matched economies, $\Delta\tilde M_j = 0$ for $j \le 2$ and $\Delta\tilde M_3 = 4\tau p_- p_+(\Delta\eta)^3$ by (17); for the K-cohort Gauss and Gauss–Radau economies with rates $\bar\eta + \Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K-2$ exactly, so $\Delta\tilde M_j = W(\Delta\eta)^j \sum_k(\omega_k^A (s_k^A)^j - \omega_k^B (s_k^B)^j)$ vanishes for $j \le 2K-2$ and equals $\pm W g_K (\Delta\eta)^{2K-1} \ne 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta\tilde M_j| \le 2W(\Delta\eta)^j$, and analyticity of ϕ in η on a disk of radius $\varrho > 0$ around $\bar\eta$, uniformly on the circle, gives $|\partial_\eta^j\phi|/j! \le C\varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega |u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}\big(z^{-1}H_0\,\partial_\eta^{j_0}\phi(\cdot,\bar\eta)\big)/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function c does not vanish identically: $f = z^{-1}H_0\partial_\eta^j\phi$ is analytic outside the disk including at infinity, where it vanishes, and is not identically zero; an analytic function on the exterior whose real part vanishes on the circle is an imaginary constant, which here would have to be zero. For $|u| \le \frac{1}{2}$, $u^2/4 \le u - \log(1+u) \le u^2$, so $D \asymp \int c(\omega)^2 d\omega\, \epsilon^2$, which gives (i)–(iii). The sample requirements need a finite-T bound that is uniform in the separation, which we obtain directly. The first T observations are $N(0, \Sigma_j)$ with Σ_j the $T \times T$ Toeplitz matrices of the spectral densities $f_j = \sigma^2|H_j(e^{i\omega})|^2$. On the compact, uniformly stable family the transfer functions have no zeros or poles on the circle (Theorem 5.7), so $0 < \underline{f} \le f_j \le \bar f < \infty$ with constants that do not depend on the

separation. A Toeplitz matrix has eigenvalues within the range of its symbol, so $\lambda_{\min}(\Sigma_0) \geq \underline{f}$ and $\|\Sigma_1 - \Sigma_0\| \leq \sup_\omega |f_1 - f_0| = \sup_\omega f_0|u| \leq \bar{f} \sup_\omega |u|$. With $A = \Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}$, whose eigenvalues lie in $[\underline{f}/\bar{f}, \bar{f}/\underline{f}]$, the inequality $\lambda - 1 - \log\lambda \leq (\lambda-1)^2/\lambda$ gives

$$D_{\mathrm{KL}}(P_1^T \| P_0^T) = \tfrac{1}{2}\sum_k (\lambda_k - 1 - \log\lambda_k) \leq \frac{\bar{f}}{2\underline{f}}\,\|A - I\|_F^2 \leq \frac{\bar{f}}{2\underline{f}}\,T\Big(\frac{\bar{f}}{\underline{f}}\Big)^2 \sup_\omega |u|^2 \leq C\,T\epsilon^2,$$

with C independent of T and of the separation, because $\sup_\omega |u| = O(\epsilon)$ uniformly. The Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6) then gives the stated lower bounds on T for every T, not only asymptotically. For a fixed pair, $T^{-1}D_{\mathrm{KL}}(P_1^T\|P_0^T) \to D$ and Stein's lemma (one error fixed) and the Chernoff bound (both errors vanishing) give the error exponents of the likelihood-ratio test, of the same order in the separation; that statement is pointwise in the pair.

Theorem 5.7. In the delay variable $\lambda = z^{-1}$, $H = \kappa\prod_i(1 - r_i\lambda)/\big[(1 - a\lambda)\prod_i(1 - r_i\lambda) - \kappa\lambda\sum_i w_i\prod_{j\neq i}(1-r_j\lambda)\big]$, which is (4) divided through by z^{K+1}; its nonstructural zeros are the memory rates, which do not cancel under the nondegenerate assumptions of Theorem 5.7. In the z-plane there is also the structural timing zero at zero: the numerator is proportional to $z\prod_i(z - r_i)$. Stability puts the poles inside the disk. A rational spectral density $\sigma^2|H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is H; Theorem 3.1 then gives the merged measure.

Proposition 6.1. With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only cohort i, so $\tilde{u}_{i,t} = r_i\tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t(\tilde{u}_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i\sum_{s\geq 0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1}u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives Σ_i. The price is a nonnegative mixture of geometric responses to its innovation (Lemma 2.3), so $\gamma_x(s) = \sum_\ell c_\ell\lambda_\ell^s$ with $c_\ell = \sigma_\xi^2\kappa^{-2}\omega_\ell\sum_m \omega_m/(1-\lambda_\ell\lambda_m) \geq 0$; substituting, $\det\Sigma_i = w_i(\gamma_x(0)A_0 - \gamma_x(1)A_1) = w_i\sum_{\ell,m} c_\ell c_m(1-\lambda_\ell\lambda_m)/(1-r_i\lambda_m)$, each term positive. By summation by parts, $\sum_{t=1}^T Z_t e_{i,t} = \sum_{t=1}^T(Z_t - r_i Z_{t+1})\varepsilon_{i,t} + O_P(1)$. Since $(\varepsilon_{i,t})$ is i.i.d. with mean zero and variance σ_f^2 and independent of the whole stationary price process, the summands $m_t = (Z_t - r_i Z_{t+1})\varepsilon_{i,t}$ form a stationary ergodic martingale difference sequence with respect to $\mathcal{F}_t = \sigma((x_s)_{s\in\mathbb{Z}}, \varepsilon_{i,s} : s \leq t)$, with $\mathbb{E}[m_t m_t^\top] = \sigma_f^2[(1+r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)] = \Omega_i$, so the Billingsley–Ibragimov martingale central limit theorem gives (19). On a uniformly stable family the modes and weights lie in a compact set with $\lambda_\ell \leq \bar{\lambda} < 1$, so $\det\Sigma_i \geq w_{\min}(1 - \bar{\lambda}^2)\gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

Theorem S12.1. With J_θ upper triangular the eigenvalues are fixed. The stationary covariance solves $P = J_\theta P J_\theta^\top + \mathrm{diag}(s_1, s_2)$, and P_{11} is affine in s_2 with positive slope, so for each θ a unique $s_2(\theta) \geq 0$ gives $P_{11} = V$ when V exceeds the value at $s_2 = 0$. The one-step price variance is s_1 throughout, and $e_1^\top J_\theta e_2 = \theta$ gives $\sup_k \|e_1^\top J_\theta^k\| \geq |\theta|$.

Proposition 6.3. (i) $\mathcal{L}_{\text{disc}} < \mathcal{L}_{\text{price}}$ is $F + Mc < M\ell G$. (ii) T_p and T_t are decreasing in $\Delta\eta$, with $T_t \geq \bar{T}$ exactly when $\Delta\eta \leq \Delta\eta_t$ and $T_p \geq \bar{T}$ exactly when $\Delta\eta \leq \Delta\eta_p$. On $\Delta\eta \leq \Delta\eta_t$ both minima equal $\bar{T}$ and $G = 0$. On $(\Delta\eta_t, \Delta\eta_p]$, $G = \bar{T} - c_t(\Delta\eta)^{-2}$, which increases to $G^* = \bar{T} - T_t(\Delta\eta_p) > 0$. On $\Delta\eta > \Delta\eta_p$, $G = c_p(\Delta\eta)^{-(4K-2)} - c_t(\Delta\eta)^{-2} = (\Delta\eta)^{-2}\big[c_p(\Delta\eta)^{-(4K-4)} - c_t\big]$; the bracket is positive exactly when $\Delta\eta < \Delta\eta_0 = (c_p/c_t)^{1/(4K-4)}$, and $G'(\Delta\eta) = (\Delta\eta)^{-3}\big[2c_t - (4K-2)c_p(\Delta\eta)^{-(4K-4)}\big] < 0$ whenever $c_p(\Delta\eta)^{-(4K-4)} > c_t$, because $4K - 2 > 2$. So G decreases wherever it is positive beyond $\Delta\eta_p$, and is non-positive beyond $\Delta\eta_0$. Every superlevel set of G at a positive level is therefore an interval (the negative tail, where G rises back toward zero from below, never reaches a positive level), and the level $(F/M + c)/\ell > 0$ lies above $G = 0$, so the set where (i) holds is an interval contained in $(\Delta\eta_t, \Delta\eta_0)$ that contains $\Delta\eta_p$ whenever it is non-empty. (iii) The interval is non-empty if and only if the level is below the maximum, $F/M + c < \ell G^*$, that is $M > M^*$; the level decreases in M, so the superlevel set grows. □

S2 Notation

Symbol	Meaning
x_t	pricing error (price minus fundamental)
a	price persistence (anchor), $0 \leq a < 1$
κ	price impact per unit of executed quantity
$q_i,\ \eta_i,\ r_i$	cohort weight, adaptation rate, memory rate $r_i = 1 - \eta_i$
$w_i = q_i\eta_i$	merged weight
$M = \sum_i w_i\delta_{r_i}$	merged rate–weight measure
$k_m,\ G(z)$	feedback kernel and its transform
$H(z),\ h_j$	closed-loop transfer function and impulse response, $h_0 = \kappa$
$J,\ S$	state transition matrix and $S = a + \kappa\sum q_i$
$v_t,\ V$	executed probe quantity and total quantity $\sum \lvert v_t\rvert$
d_T	impulse-response difference between hypotheses, first T taps
σ	observation-noise standard deviation
$\delta,\ \Delta\eta$	half-separation of a known pair; spacing of unknown cohorts

Symbol	Meaning
K_*	transfer-difference constant $2\kappa^2 q/[\epsilon^3(1-S)^2]$
T_h	Toeplitz matrix $(h_{\lvert i-j\rvert})$ of execution cost
$g_i(\omega)$	dissipation kernel $\mathrm{Re}\,\hat{H}_i(e^{i\omega}) - \kappa/2$
$B,\ \mu,\ c,\ \phi$	capital, maintenance-margin fraction, proportional cost, taker fee
V_{eff}	per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$
$\hat{Q},\ \hat{S}$	reduced-form loop gain and persistence indices

S3 Further results

S3.1 A finite-window price-surveillance bound

Theorem S3.1 (Finite-window abstention)**.** *Observe $x_t = sx_{t-1}+w_t$, $w_t \sim N(0,\sigma^2)$, $x_0 = 0$, $t \le T$, with $s \in (0,2)$. Let a possibly randomized rule Ψ output Stable, Unstable or Abstain, with $\sup_{s<1} P_s(\Psi = \text{Unstable}) \le \alpha$ and $\sup_{s\ge 1} P_s(\Psi = \text{Stable}) \le \alpha$, $\alpha < 1/2$. Choose $0 < c_0 < 1$ with $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. On $B_T = (1 - c_0/T, 1 + c_0/T)$,*

$$\inf_{s\in B_T} P_s(\Psi = \text{Abstain}) \ \ge\ 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0,$$

and for every finite T the minimax worst-case error of binary rules over $s \in (0,2)$ is at least $1/2$.

Proof. Pair $s_0 \in (1-c_0/T, 1)$ with $s_1 = s_0 + c_0/T$. Then $D_{\mathrm{KL}}(P_{s_0}\|P_{s_1}) = \frac{(s_1-s_0)^2}{2\sigma^2}\sum_t \mathbb{E}x_{t-1}^2 \le c_0^2 e^{2c_0}/4$, so Pinsker bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Stable}) \le \alpha + \text{TV}$ and $P_{s_0}(\text{Unstable}) \le \alpha$, which yields the abstention bound. For binary rules, continuity in total variation and Le Cam's two-point bound give worst-case error tending to $1/2$. □

The interval B_T is an abstention layer that any error-controlled monitor must respect in the scalar Gaussian case; richer disclosure can remove it.

S3.2 Passive distinction of a known pair in finite samples

Proposition S3.2 (Passive distinction with zero controlled quantity)**.** *Keep the known pair, but add independent uniform innovations $w_t \in [-b, b]$ of variance $s_w^2 = b^2/3$ at each price update, observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0,\sigma^2)$, use independent three-step reset blocks and*

no probes, with $P_* > 3b$. Let $b_1 = a + \kappa q\eta$, $b_{2,0} = b_1^2 + \kappa q\eta(1-\eta)$ and $b_{2,\delta} = b_{2,0} - \kappa q\delta^2$. In each block the covariance of the first and third observations differs by $\Delta = \kappa q s_w^2 \delta^2$. With

$$R = \max_{r\in\{0,\delta\}} \Big\{ (s_w^2+\sigma^2)\big[s_w^2(1+b_1^2+b_{2,r}^2)+\sigma^2\big] + s_w^4 b_{2,r}^2 \Big\},$$

the midpoint test of the average product has worst-case error at most α whenever $n \geq 4R/(\alpha\Delta^2)$ blocks are observed: zero controlled quantity and order δ^{-4} blocks.

The Chebyshev bound is conservative but the order is real. With $b = 0.03$, observation s.d. 0.01 and $\delta = 0.10$, the bound asks for 3.9×10^7 blocks at $\alpha = 0.1$; in simulation the midpoint test still errs 32–38% of the time with 2×10^5 blocks. Independent resets and known innovation laws are extra conditions. The proposition does not show that anonymous public records identify participants; it shows that passive excitation can accumulate memory information while the probe account stays untouched.

Proof. In each block $x_1 = w_1$, $x_2 = b_1 w_1 + w_2$ and $x_3 = b_{2,r} w_1 + b_1 w_2 + w_3$, so $|x_t| \leq 3b$. Observation noise changes variances but not the covariance $s_w^2 b_{2,r}$. The product variance is the product of marginal variances plus the squared covariance plus a fourth-cumulant term $b_{2,r}^2(\mathbb{E}w^4 - 3s_w^4) \leq 0$ for uniform innovations, so it is at most R. Chebyshev bounds each midpoint error by $4R/(n\Delta^2)$. □

S3.3 Blind inversion of the closed loop

1. *Impulse decoupling:* $g_t = h_{t+1} - a h_t$.
2. *Minimal realization:* form the Hankel matrix $H_{ij} = g_{i+j-1}$ and run Kung's eigensystem realization with a truncated SVD.
3. *Transmission zeros:* compute the zeros of the realization, which under known (κ, a) isolate the roots of the denominator of G.
4. *Weights:* with recovered rates, project $u_t = (x_t - a x_{t-1})/\kappa$ on the reconstructed filters $y_{k,t-1}$.

Without noise each step is exact: the realization has the order of the true system, its nonstructural transmission zeros are the memory rates (Theorem 3.1); the known timing zero is discarded, and the projection returns the weights, so the procedure removes the pole-shift bias of fitting poles directly to the price response. With noise, per-cohort recovery degrades as rates move together at the rate given by Theorems 5.3 and 5.4, which is why we do not report it as an estimator of individual cohorts.

S3.4 Probing protocols

Table S2 compares four probing protocols at observation horizon $T = 4$ per cycle.

Table S2: Probing protocols at observation horizon $T = 4$ per cycle

Protocol	δ	Executed vol.	Max trade	Error (SE)	Capital
A: single pulse (unconstrained)	0.03	131.5	131.5	0.047 (0.003)	6,985.91
	0.05	47.3	47.3	0.054 (0.004)	921.73
	0.08	18.5	18.5	0.053 (0.004)	146.73
	0.12	8.2	8.2	0.044 (0.003)	31.45
B: repeated capped pulses	0.03	2,768.8	6.2	0.050 (0.003)	7,035.75
	0.05	362.5	6.2	0.054 (0.004)	923.88
	0.08	56.2	6.2	0.047 (0.003)	146.00
	0.12	12.5	6.2	0.038 (0.003)	34.88
C: sequential test, capped	0.03	1,410.2	6.2	0.045 (0.003)	3,585.14
	0.05	195.6	6.2	0.038 (0.003)	500.00
	0.08	35.9	6.2	0.032 (0.003)	94.30
	0.12	10.5	6.2	0.011 (0.002)	29.85
D: generalized eigenvector	0.03	2,359.7	3.7	0.050 (0.003)	2,632.77
	0.05	314.8	3.7	0.048 (0.003)	353.67
	0.08	49.2	3.7	0.046 (0.003)	57.53
	0.12	9.9	3.8	0.045 (0.003)	13.56

2,000 replications per hypothesis and cell. Log–log capital slopes: -3.90, -3.84, -3.47, -3.81. Protocols A and B reproduce the earlier version exactly. C and D are re-implemented because the earlier code was not retained: C is a Wald test with capped pulses; D repeats the top generalized eigenvector of a flat $(T + 1)$-step probe, scaled to a maximum trade of 3.75, and is cheaper than the earlier D.

S3.5 Coordinate-weighted output

For $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$ with $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.3$ ($\rho(J) = 0.8224$, non-normal), the price output has $\sup_t \|e_x^\top J^t\| = 1$. The weighted output $C_w = (1, \lambda/\mathrm{MMR}) = (1, 160)$ with $\lambda = 0.8$ and $\mathrm{MMR} = 0.005$ has $\sup_t \|C_w J^t\| = \|C_w\| = 160.003$, attained at $t = 0$, then 122.32, 98.32, 80.79 . The factor 160 is a static coordinate weight; $C_w z$ is not a margin quantity because z contains no inventory. Margin must be computed on a cash-and-inventory ledger.

S4 Extended related literature

Algorithms, AI and supervision. Recent work in finance asks how learning algorithms and AI agents behave in markets: reinforcement-learning market makers and informed traders can sustain wide spreads or collusive outcomes (Colliard et al., 2022; Dou et al., 2025), prices formed against Q-learners have distinctive dynamics (Banerjee and Szydlowski,

2026), and algorithmic markets can be manipulated (Tremacoldi-Rossi, 2025). Traders who differ in horizon or speed are studied in equilibrium by Cespa and Vives (2015) and Biais et al. (2015), and Crouzet et al. (2020) model investors who trade at different frequencies. These papers model or simulate the algorithms. We take the supervisor's side and ask what can be learned about them from market data, and what disclosure is worth. A related literature measures the cost of regulation and of regulatory reporting (Trebbi et al., 2024); our dollar comparison with the Consolidated Audit Trail is a statement about the benefit side: what the audit trail buys that prices cannot.

Price formation and impact. Our price equation is a discrete linear impact model in the tradition of Kyle (1985), Glosten and Milgrom (1985) and Hasbrouck (1991), in which order flow moves prices and prices revert toward value. Transient impact and the cost of trading follow Almgren and Chriss (2001), Obizhaeva and Wang (2013), Gatheral (2010) and Bouchaud et al. (2004); Gatheral (2010) shows that absence of dynamic arbitrage requires a positive-definite decay kernel, Alfonsi et al. (2012) that excluding transaction-triggered manipulation requires convex, nonincreasing decay, and Alfonsi et al. (2016) extend positive definiteness to several assets. Positive definiteness is the property Lemma 4.2 delivers in our cohort economy. Bouchaud et al. (2018) document order splitting and long memory in order flow, which is the main confounder in Supplement S13. Relative to this literature we treat the impact kernel not as a primitive but as the closed-loop output of adaptive traders, and ask what it reveals about them.

Algorithmic trading and feedback. Algorithms that react to recent prices can amplify or dampen moves. Brunnermeier and Pedersen (2009) and Shleifer and Vishny (1992) study liquidity spirals and fire sales, and Kirilenko et al. (2017) document the behaviour of automated intermediaries in the 2010 Flash Crash. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles. Our contribution is on the measurement side: we show what a closed-loop observer can and cannot learn about the feedback, and that standard reduced-form indices are dominated by filtering and timing.

Learning algorithms and what can be inferred about them. Empirical game-theoretic analysis treats the choice among simulated trading strategies as a game (Wellman, 2006). Empirically, algorithmic trading has been found to improve liquidity and price discovery on average (Hendershott et al., 2011; Brogaard et al., 2014). These studies simulate the algorithms or observe them through regulatory data; we ask what an outside observer can infer about them from market data alone, and answer in samples and in capital. Our

stance is that of partial identification (Manski, 2003): when the structure is not identified, characterize what the data do determine, here the merged measure and, through it, the intensity-weighted distribution of the desks' information quality (Proposition S5.1).

Control without identification. We do not claim that acting on a system without identifying it is new. Moallemi et al. (2008) study strategic execution against an observer who learns from prices, which bears directly on our assumption that cohorts do not react to being observed, and Park and Van Roy (2015) study execution that learns price impact while trading, with regret guarantees; our object is different, the latent trader cohorts behind the impact rather than the aggregate impact coefficients. Active identification of linear systems has matching bounds and optimal experiment design (Wagenmaker and Jamieson, 2020), and task-directed exploration asks what must be learned for a decision (Wagenmaker et al., 2021); our information matrix differs because the unknowns are zeros of a closed-loop response and the input is paid for in trading losses. van Waarde et al. (2020) characterize when control is possible without unique system identification. Our results concern a specific market observation contract and its costs in executed quantity, capital and money.

Closest work. The closest literature is super-resolution and moment-based mixture estimation, which gives rates for recovering clustered point sources and mixing measures from noisy moments (Moitra, 2015; Heinrich and Kahn, 2018; Batenkov et al., 2021; Liu and Ammari, 2022; Wu and Yang, 2020; Kong and Valiant, 2017). Our K-cohort rate is, in that language, the square of the amplitude condition number $(\Delta\eta)^{-(2K-1)}$ for K clustered nodes (Heinrich and Kahn, 2018; Wu and Yang, 2020), and we do not claim the exponent as new. What changes when the mixture sits inside a closed-loop financial market is fourfold (Table S3): first, in mixture estimation and open-loop system identification the unknown rates are locations or poles of the observed law, whereas in a closed-loop market the cohorts feed back into the price, making the memory rates transmission *zeros* of the observable transfer function $H(z)$ that must first be extracted from the loop (Theorem 3.1); with a and κ known this extraction is a smooth invertible map, so it decides which method is valid (pole fitting on prices is not) rather than the rate; second, passive price observations face at least the same $(\Delta\eta)^{-(4K-2)}$ sample-complexity barrier as active probing, uniformly in the spacing, because $H(z)$ is minimum phase and the Whittle likelihood inherits the moment gap (Theorems 5.7 and 5.8); third, in a market statistical sample size translates into executed quantity $\Theta(\delta^{-2})$ and round-trip trading loss or margin capital $\Theta(\delta^{-4})$ via the self-impact Toeplitz quadratic form $\frac{1}{2}v^\top T_h v$ (Theorems 4.1–S9.4); and fourth, class-tagged disclosure breaks the feedback loop cohort by cohort and cuts the sample requirement by $(\Delta\eta)^{-(4K-4)}$

(Proposition 6.1 and Theorem 6.2). Beyond the K-cohort case, we prove matching lower and upper bounds of exact logarithmic order $\Theta(1/\log N)$ for general completely monotone memory (Theorem 5.6) and price the resulting disclosure savings against audit-trail budgets. Several ingredients are known in other settings, and we claim only their closed-loop, passive and financial forms: the δ^{-4} law for telling one component from two closely spaced ones is classical in resolution analysis (Shahram and Milanfar, 2005); logarithmic rates are known for mixtures of exponentials (Rebafka and Roueff, 2015) and for general mixing distributions, by the same moment-matching and polynomial-approximation technique (Wu and Yang, 2020); and labelled samples are known to be far more informative than unlabelled ones in mixture problems (Castelli and Cover, 1996).

Estimating heterogeneous-agent models from prices. A large literature models markets as populations of trader types that switch between forecasting rules (Brock and Hommes, 1998) and estimates such models from price data. Estimates of the parameters that separate the types are often imprecise: Amilon (2008) reports that the estimated model reproduces properties of returns but that its fit is generally poor, Franke and Westerhoff (2012) compare switching mechanisms by the method of simulated moments, and Kukacka and Barunik (2017) find the switching coefficient statistically insignificant by simulated maximum likelihood while the belief parameters are significant. These papers take the identification of the types from prices as given. Our results suggest one reason the type-separating parameters are hard to pin down: around a steady state, prices determine only the merged measure, and separating types whose memories differ by $\Delta\eta$ takes a number of samples of order $(\Delta\eta)^{-(4K-2)}$ for K types (Theorems 3.1 and 5.3). The models these papers estimate are nonlinear, so this is a statement about their linearization, not a claim that their estimates are wrong; it does give a quantity, the separation of the types' memories, that bounds the precision one can expect before estimating.

Closed-loop system identification. Engineering has long studied identification under feedback (Ljung, 1999; Forssell and Ljung, 1999). There the object is the plant, the difficulty is that feedback correlates the input with the noise, and the remedy is external excitation that makes the experiment informative. Our problem is the reverse in two ways. The object is the feedback law (the traders), not the plant (the price), and the memory rates enter as zeros of the observed transfer function. And external excitation does not remove the difficulty: with a probe the merged measure is identified exactly (Theorem 3.1), so the barrier is not qualitative identifiability but the conditioning of the map from memories to responses, which we measure in samples, quantity and capital. In finance, Leach and Madhavan (1993) show

Table S3: Comparison with finite-mixture estimation and open-loop identification

Feature	Finite-mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020)	Open-loop exponential fitting (Roy and Kailath, 1989; Batenkov et al., 2021)	Closed-loop market identification (this paper)
Where rates live	Component locations / moments of density	Poles of impulse response	Transmission zeros of closed-loop $H(z)$ (Thm 3.1)
Active vs. passive data	i.i.d. draws only	Designed input vs. output spectrum	At least the same $(\Delta\eta)^{-(4K-2)}$ order, uniformly, via the minimum-phase Whittle gap (Thm 5.8)
Economic unit of cost	Sample size N	Signal-to-noise ratio	Quantity $\Theta(\delta^{-2})$, trading loss and capital $\Theta(\delta^{-4})$ via $\frac{1}{2}v^\top T_h v$ (Thms 4.1–S9.4)
Effect of class tagging	Observed component labels ($\Theta(1)$)	Separate sensor per mode	Breaks closed loop cohort-by-cohort; saves $\Theta((\Delta\eta)^{-(4K-2)})$ at fixed accuracy, $\Theta((\Delta\eta)^{-(4K-4)})$ for rates (Thm 6.2)

that a market maker can gain by setting prices to learn about demand, so learning through trading has an equilibrium counterpart; our probing observer is a reduced-form version of that idea without the equilibrium. In econometric terms, as $\Delta\eta \to 0$ the cohort parameters pass from strong to weak identification (Andrews and Cheng, 2012), and our rates state that passage in samples.

Inferring trader types from market data. Market microstructure has a tradition of inferring hidden trader populations from public data. The probability of informed trading (Easley et al., 1996) fits a mixture model to daily counts of buys and sells to estimate the share of informed orders, and what it captures has been debated (Duarte and Young, 2009). With regulatory account data, Kirilenko et al. (2017) classify traders by their holding and trading patterns. Our results speak to the case where types differ in how long they remember prices: from prices alone the separation is costly, and account or class records of the kind Kirilenko et al. (2017) use remove most of the cost (Proposition 6.1).

Statistical identification. The identification results use two-point testing arguments (Le Cam, 1986; Tsybakov, 2009). The moment-matching construction of Section 5 is the dynamical analogue of the slow rates in finite-mixture estimation (Manole and Ho, 2022) and of super-resolution of near-colliding sources (Liu and Ammari, 2022; Batenkov et al., 2021); the

Table S4: Closest results and how this paper differs

Existing result	Observations	Target	Conclusion	Difference from this paper
Mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020)	i.i.d. draws from a K-component mixture	Mixing measure	Rates that slow sharply as components merge; $n^{-1/(4K-2)}$ for Gaussian location mixtures	Same exponent; here the measure is observed through a feedback loop, as zeros of H, actively or passively, and the cost is also priced in quantity and capital
Super-resolution and Prony (Moitra, 2015; Batenkov et al., 2021; Katz et al., 2023)	Noisy Fourier or exponential samples	Point-source locations	Stability governed by Vandermonde conditioning and node separation; Prony attains the min-max bound for clustered exponents	Real nodes extracted from a closed loop; our matching upper bound (Theorem 5.4) adapts this known result
Closed-loop system identification (Ljung, 1999; Forssell and Ljung, 1999)	Input–output data under feedback	The plant	Consistency requires informative (externally excited) experiments	Target is the feedback law (the traders); excitation gives exact identification of M, and the barrier is conditioning, measured in samples and capital
Heterogeneous-agent estimation (Brock and Hommes, 1998; Franke and Westerhoff, 2012; Kukacka and Barunik, 2017)	Prices and returns	Type and switching parameters	Type-separating parameters often imprecise	Bounds that precision by the types' memory separation (in the linearization)
Probability of informed trading (Easley et al., 1996)	Daily buy and sell counts	Share of informed orders	Mixture estimate of trader population	Types that differ in memory; price-only versus class-tagged data
Data informativity (van Waarde et al., 2020)	Finite data, often noiseless	Whether a property is determined	Qualitative conditions	Quantitative minimax costs under noise

general-memory result combines best-approximation duality (Bernstein, 1914; Rivlin, 1981) with the moment comparisons used by Kong and Valiant (2017) and Wu and Yang (2020), and its logarithmic rate parallels supersmooth deconvolution (Fan, 1991); recovery of exponentials connects to ESPRIT and matrix-pencil methods (Roy and Kailath, 1989; Hua and Sarkar, 1990). Minimum-phase spectral factorization and the Whittle likelihood are classical (Whittle, 1953). What is new here is the mapping from these statistical rates into executed quantity and money through a financially specified account, the passive equivalence, and the value of class records; that the relevant rates are zeros, not poles, of the observable transfer function matters for which estimator is valid, not for the rate.

Non-normal dynamics and supervision. Transient growth of stable non-normal systems is classical (Trefethen and Embree, 2005; Kreiss, 1962). We use it to show that supervision through eigenvalues and second moments of prices is blind to transient risk (Theorem S12.1), which motivates the identification-free control of the companion paper. On the regulatory side, pre-trade risk controls for broker-dealers with market access are governed by SEC Rule 15c3-5 (SEC, 2010), and participant-level audit trails are the natural disclosure regime for Proposition 6.1. A large literature studies how pre- and post-trade transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006); we ask a different question, what disclosure lets a supervisor learn about the algorithms that trade, and give the answer in samples. Crypto venues, whose market structure is studied by Makarov and Schoar (2020), are a natural laboratory for this question because they publish class-level positioning every five minutes, while the CFTC publishes it weekly for regulated futures.

Partial fractions, Gaussian testing, convolution inequalities and positive-system gains are established tools. Our contribution is to connect them into a quantitative chain from memory resolution, through probing resources, to money, with matching orders, explicit constants and a supervisory interpretation, and then to measure the relevant quantities in data.

S5 What prices hide, in economic terms

The worked example states the barrier in signal-to-noise ratios. That is the general picture. Write $s = \sigma_\zeta^2/\sigma_\nu^2$ for a desk's signal-to-noise ratio, so that by (7) its memory rate is $1 - \eta(s)$ with

$$\eta(s) = \tfrac{1}{2}\big(\sqrt{s^2+4s} - s\big), \qquad \eta'(s) = \tfrac{1}{2}\Big(\frac{s+2}{\sqrt{s^2+4s}} - 1\Big) > 0,$$

and η is increasing and concave. With no inventory penalty, the first-order condition of the Remark on exponential memory gives the trading intensity $q_i = \beta_i/\lambda_{\mathrm{tc},i}$: what the desk expects to earn per unit of belief, over what it pays to trade.

Proposition S5.1 (Identification in economic primitives)**.** *Consider K Kalman desks (Lemma 2.1(i)) with signal-to-noise ratios $s_i \in [\underline{s}, \bar{s}] \subset (0, \infty)$ and intensities $q_i = \beta_i/\lambda_{\mathrm{tc},i}$, and write $c_- = \eta'(\bar{s})$ and $c_+ = \eta'(\underline{s})$.*

(i) Prices and aggregate order flow depend on the desks only through the measure

$$\sum_i q_i \, \eta(s_i) \, \delta_{1-\eta(s_i)}.$$

Desks are identified through their signal-to-noise ratios and intensities only; expected returns and trading costs enter only through their ratio, and two populations with the same intensity-weighted distribution of signal-to-noise ratios are observationally equivalent.

(ii) Fix a centre s_c and let the ratios lie in $s_c + \Delta s\,[-1, 1] \subset [\underline{s}, \bar{s}]$. Recovering the intensity-weighted distribution of signal-to-noise ratios to Wasserstein accuracy of order Δs, or its weights $q_i\eta(s_i)$, from N noisy reset experiments with worst-case error probability below $1/4$ requires $N \geq c\,\sigma^2(\Delta s)^{-(4K-2)}$; on separated configurations, Prony's method of Theorem 5.4 composed with η^{-1} achieves this order. The constants are those of Theorems 5.3 and 5.4 multiplied by powers of c_- and c_+.

In words. What prices reveal about Kalman-filter desks is the mix of how good their information is (their signal-to-noise ratios), weighted by how hard they trade. Expected returns and trading costs matter only through their ratio. Pinning down that mix carries the same steep cost as separating memory rates.

Proof. (i) By Lemma 2.1(i) desk i is a cohort with memory rate $1 - \eta(s_i)$ and weight q_i, so Theorem 3.1 applies, and $\beta_i, \lambda_{\mathrm{tc},i}$ enter the economy only through q_i. (ii) On $[\underline{s}, \bar{s}]$, η is a C^1 bijection onto its image with $c_- \leq \eta' \leq c_+$. Lower bound: take the two economies of Theorem 5.3 with centre $r_c = 1 - \eta(s_c)$ and rate spacing $\Delta\eta = c_-\Delta s$. By the mean value theorem their gains are images of ratios in $s_c + \Delta s\,[-1, 1]$, so they are two populations of Kalman desks in the class, and the bound $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)} = c\,c_-^{-(4K-2)}\sigma^2(\Delta s)^{-(4K-2)}$ applies to recovering their weights. Because η is c_+-Lipschitz, the Wasserstein distance between the two distributions of ratios is at least $1/c_+$ times that between the two distributions of rates, which Theorem 5.3 bounds below by a constant times $\Delta\eta = c_-\Delta s$. Upper bound: ratios in

$s_c + \Delta s\,[-1, 1]$ with spacing at least $\gamma\Delta s$ have gains in $\eta(s_c) + c_+\Delta s\,[-1, 1]$ with spacing at least $c_-\gamma\Delta s$, a class of the form required by Theorem 5.4 with rescaled separation $\gamma c_-/c_+$; Prony's method recovers the rates and weights at that theorem's rate, and η^{-1}, which is $1/c_-$-Lipschitz, maps rate errors to ratio errors. □

The private information a supervisor would most like to know, how good each desk's information is, is exactly what prices hide. In the worked example the two desks' ratios differ by a factor of almost two, and separating them already costs hundreds of thousands of dollars of probing capital. The proposition takes intensities to vary independently of information quality, because they also reflect costs and capital. The next proposition asks what happens when the economics ties them together.

Proposition S5.2 (Intensity tied to information quality)**.** *Let each desk's intensity be a known function of its signal-to-noise ratio, $q = \varphi(s)$ with φ positive and C^2 on $[\underline{s}, \bar{s}]$ (for a mean–variance desk, $\varphi(s)$ increases with s), and let $n_i \geq 0$ denote the mass of desks with ratio s_i (a divisible population, not an integer desk count). Write $s(\eta)$ for the inverse gain map and $w(\eta) = \varphi(s(\eta))\,\eta$ for the weight per desk as a function of its gain.*

(i) Unknown numbers of desks. *The merged measure is $\sum_i n_i\, w(\eta(s_i))\,\delta_{1-\eta(s_i)}$. Every positive measure with finitely many atoms in $[1-\eta(\bar{s}), 1-\eta(\underline{s})]$ is of this form, with $n_i = m_i/w(\eta_i)$ for an atom of mass m_i at $1-\eta_i$. So when the numbers of desks of each type are not known, tying intensity to information quality removes no economy, and every lower bound of Proposition S5.1(ii) holds unchanged.*

(ii) Known numbers of desks. *If the n_i are also known, the set of economies is smaller, but the single-versus-pair problem keeps its order: one type at gain $\bar{\eta}$ with $2n$ desks and two types at $\bar{\eta} \pm \delta$ with n desks each have taps that differ by $n\,\delta^2\,\partial_\eta^2[w(\eta)(1-\eta)^m]_{\bar{\eta}} + o(\delta^2)$, so separating them needs $N \geq c\,\sigma^2\delta^{-4}$ reset experiments, the order the symmetric pair has when intensities vary freely.*

(iii) Self-confirming configurations. *Suppose the ratio each desk perceives is the one implied by the closed-loop dynamics it trades in, through a map T from configurations $(s_1, \ldots, s_K)$ to perceived ratios that is continuous on a box $\prod_i[\underline{s}, \bar{s}]$ and maps it into itself. Then a configuration with $T(s) = s$ exists, and parts (i)–(ii) and Proposition S5.1 apply at it.*

Proof. (i) Desk type i is a cohort with gain $\eta_i = \eta(s_i)$, memory rate $1-\eta_i$ and total weight $n_i\varphi(s_i)$, so it contributes $n_i\varphi(s_i)\eta_i = n_i w(\eta_i)$ to the atom at $1-\eta_i$. Since $w > 0$, any atom mass $m_i \geq 0$ is attained with $n_i = m_i/w(\eta_i)$. (ii) Write $g_m(\eta) = w(\eta)(1-\eta)^m$; the tap

difference is $n[g_m(\bar{\eta}+\delta)+g_m(\bar{\eta}-\delta)-2g_m(\bar{\eta})] = n\delta^2 g_m''(\bar{\eta})+o(\delta^2)$, with $\sum_m \sup|g_m''|^2<\infty$ on gains bounded away from zero, so the Euclidean distance between the two tap vectors is $O(n\delta^2)$ and the Kullback–Leibler divergence over N experiments is $O(Nn^2\delta^4/\sigma^2)$; Le Cam's two-point argument gives the bound. (iii) is Brouwer's fixed-point theorem. □

So the barrier does not depend on intensities varying freely: it survives any known link between intensity and information quality as long as the supervisor does not also know how many desks of each kind there are, and even then the simplest separation problem keeps its δ^{-4} order. Whether the full $(\Delta\eta)^{-(4K-2)}$ exponent survives with known numbers of desks is open. Part (iii) shows only that self-confirming configurations exist under continuity; it does not derive the map T from a model of the desks' learning.

S6 Natural excitation and the cost of ambiguity

Natural price innovations can also separate the pair with zero controlled quantity: an exact finite-sample test on independent three-step blocks needs order δ^{-4} blocks (Online Supplement, Proposition S3.2), the same order as the passive Whittle bound of Section 5. Non-identification also has a price on the quote side: when the relevant set of economies includes costlier ones, a robust dealer widens its quote, which the companion paper analyses (Niroula et al., 2026); cohort non-identification alone does not establish that the worst relevant cost exceeds the informed one.

S7 Local information: the Cramér–Rao benchmark

Proposition S7.1 (Cramér–Rao benchmark)**.** *Let θ collect the weights $w_1,\dots,w_K$ and rates $\eta_1,\dots,\eta_K$. Suppose the impulse response $h_t(\theta)$, $t\le T$, is observed with i.i.d. Gaussian error from N reset probes of size V_{probe} against innovation noise σ_p, so the error has standard deviation $\sigma_h=\sigma_p/(V_{\mathrm{probe}}\sqrt{N})$. With Jacobian $J_{tj}=\partial h_t/\partial\theta_j$ and $F_{\mathrm{unit}}=J^\top J$, any regular unbiased estimator has* $\mathrm{Var}(\hat{w}_k)\ge\sigma_h^2[F_{\mathrm{unit}}^{-1}]_{kk}$*, and a 95% interval of half-width* tol *requires*

$$N^*\ge\Big(\frac{1.96}{\mathrm{tol}}\Big)^2\Big(\frac{\sigma_p}{V_{\mathrm{probe}}}\Big)^2[F_{\mathrm{unit}}^{-1}]_{kk}. \tag{S1}$$

For $K=2$ with spacing $\Delta\eta$, $[F_{\mathrm{unit}}^{-1}]_{kk}=\Theta((\Delta\eta)^{-6})$, so $N^*=\Theta((\Delta\eta)^{-6})$.

In words. This gives the number of test trades any well-behaved estimator needs to pin down one group's trading weight to a stated precision. Each extra group with a nearby

Table S5: Cramér–Rao sample sizes for multi-cohort weight recovery ($T = 120$ taps)

Scenario	K	N^* probes	σ_{crit}	ML success ($f = 1$)	RMSE/CRB
$\eta = 0.30$	1	0.1	6.05×10^{-3}	94%	1.02
$\Delta\eta = 0.30$	2	47.8	2.31×10^{-4}	94%	1.01
$\Delta\eta = 0.20$	2	597	6.55×10^{-5}	96%	0.98
$\Delta\eta = 0.10$ ($\delta = 0.05$ pair)	2	40,374	7.96×10^{-6}	93%	1.03
$\Delta\eta = 0.05$	2	2,620,320	9.88×10^{-7}	95%	1.03
spacing 0.10	3	17,349,791	3.84×10^{-7}	98%	1.22
spacing 0.20	3	27,038	9.73×10^{-6}	98%	0.88

Tolerance ± 0.01 at 95%. σ_{crit} is the single-probe noise at which one probe suffices; operational single-probe noise is 1.6×10^{-3}. ML columns: least squares over all weights and rates from six generic starts (never the true values), 200 replications per row at σ_{crit}; success means every weight within ± 0.01. Fitted log–log slope of N^* in $\Delta\eta$: -6.04.

memory raises the number sharply, as Table S5 shows.

At the critical noise level maximum likelihood attains the bound: its RMSE for the hardest weight is between 0.88 and 1.22 times the Cramér–Rao value and every weight is within tolerance in 93% to 98% of replications (Table S5), so the critical noise levels, not estimator failure, are what make close cohorts expensive. Under ordinary surveillance budgets ($N \leq 50$) unbiased estimators cannot resolve close weights.

S8 A hypothetical scale illustration: disclosure in dollars

This subsection is a hypothetical scale illustration of an outside prober's shadow loss, not a public cost–benefit calculation.

Theorem 6.2 states the value of class-tagged disclosure in samples. Theorems S9.2 and 5.3 let us translate that value into a flow of round-trip trading losses for an outside participant. Two adjustments make the figures exact:

(i) *Exact K-cohort moment-gap constant.* Rather than extrapolating from Cramér–Rao ratios, we evaluate the spectral round-trip trading loss bound of Theorem S9.2,

$$L^*_{\text{rt}}(K, \Delta\eta) \;=\; \frac{4\sigma^2 z^2_{1-\alpha}}{\sup_\omega \big(|\Delta\hat{H}_{K,\Delta\eta}(e^{i\omega})|^2 / g_0(\omega)\big)},$$

directly for the K-point Gauss versus Gauss–Radau economies of Theorem 5.3 and Table 6, whose moments below order $2K-1$ coincide and whose $(2K-1)$-th moment gap has the exact constant $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, and $\Delta_4 = 0.013061$ at spacing $\Delta\eta$ (with $\Delta\eta = 2\delta$ for the symmetric known pair).

(ii) *Flow against flow.* A single probe loss L^*_{rt} is a one-time certification cost, whereas an audit trail is an ongoing annual expenditure. Because algorithm populations turn over, we report the annualised round-trip trading loss $52 \times L^*_{\mathrm{rt}}$ under weekly renewal (matching the CFTC weekly reporting cycle of Section 6.3). For orientation only, we also record official audit-trail figures of two different kinds; we do not divide one by the other, because a prober's trading loss is a private transfer to other traders calibrated to a Bitcoin perpetual, while an audit trail's budget is a resource cost for equities and options. The first is CAT LLC's own operating budget: \$248.8 million for 2025 in the November 2024 filing (SEC Release No. 34-101901, Dec. 12, 2024, p. 4), revised to \$228.3 million in May 2025 and reduced further later in 2025, and the revised 2026 Financial and Operating Budget of 31 March 2026, with total expenses of \$147,044,869, which we use (CAT LLC, 2026). The second is historical, ex ante estimates of industry-wide compliance costs, which are not operating budgets and cover a different scope: the SEC's 2016 CAT NMS Plan Approval Order estimate of \$2.4 billion in initial industry build costs and \$1.7 billion per year in ongoing industry costs (SEC, 2016, Release No. 34-79318, Nov. 15, 2016, 81 FR 84696 at 84801, p. 363); and the SEC's Rule 613 proposing and adopting releases estimating \$4.0 billion initial and \$2.1 billion annual ongoing costs (SEC, 2012, Release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153; Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238).

The figures are a shadow cost of learning by trading, the price an outside participant would pay; a supervisor without an audit trail does not probe, and its cost is detection delay, which Proposition 6.3 models. The cost results require a positive economy: with signed weights the execution form $\frac{1}{2}v^\top T_h v$ can be indefinite and a round trip can earn money, so none of the dollar figures applies there. Probing cost is a lower bound on the cost of learning from prices by trading, not on passive learning and not a social cost. It is also a lower bound for deterministic probes only: sequential probing can lower the expected loss by a factor of up to 0.49 (Theorem S9.3). Only the observation noise is estimated from data; the structural parameters are illustrative, and the constant is sensitive to them. At $\delta = 0.05$ the numerically evaluated lower bound of Theorem S9.2 is 358.8, 179.0, 61.1 and 13.9 par units for $a = 0$, 0.25, 0.5 and 0.7, and 3.9, 19.6, 61.1 and 305.2 for $\bar{\eta} = 0.2$, 0.3, 0.4 and 0.6; at $\Delta\eta = 0.20$ instead of 0.10 the unknown-pair figures fall by a factor of about 64. The Gauss–Radau pair is used in the orientation that gives the larger, still valid, lower bound. The known-pair rows span a rate width of $2\delta = \Delta\eta$, whereas the $K = 2$ Gauss–Radau rows on $r_c + \Delta\eta[-1, 1]$ span $\frac{4}{3}\Delta\eta$. Within that interpretation, Table S6 shows how rapidly the polynomial exponent $(\Delta\eta)^{-(4K-2)}$ and the shrinking Gauss–Radau moment-gap constant

Table S6: Price-only round-trip probing loss per market and annualised weekly-renewal cost (hypothetical outside prober)

	Loss / probe (USDT)		Annual (52×, USDT)	
Economy ($\Delta\eta$)	Roll	1-sec	Roll	1-sec
$K = 2$, known (0.20)	6,200	149,901	322,400	7,794,852
$K = 2$, known (0.10)	99,192	2,398,422	5,157,984	124,717,944
$K = 2$, known (0.04)	3,874,696	93,688,369	201,484,192	4,871,795,188
$K = 2$, Radau (0.20)	4,292	103,788	223,184	5,396,976
$K = 2$, Radau (0.10)	283,488	6,854,612	14,741,376	356,439,824
$K = 3$, Radau (0.20)	804,403	19,450,107	41,828,956	1,011,405,564
$K = 3$, Radau (0.10)	1,137,325,880	27,500,017,407	59,140,945,760	1,430,000,905,164
$K = 4$, Radau (0.20)	156,632,086	3,787,291,895	8,144,868,472	196,939,178,540

Separation variable $\Delta\eta$ throughout ($\Delta\eta = 2\delta$ for the symmetric known pair). Gauss–Radau rows use the exact K-cohort quadrature pair of Theorem 5.3 and Table 6 (moment gaps $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, $\Delta_4 = 0.013061$ at $W = 0.08$, $r_c = 0.60$). Loss per probe is the spectral round-trip trading loss bound of Theorem S9.2 at the median BTCUSDT Roll noise (0.40 USDT) and median one-second price-change standard deviation (1.98 USDT); annual loss multiplies by 52 weekly renewals. Probing cost is a lower bound on the cost of learning from prices by trading, not on passive learning; it is a private transfer to other traders, not a social or resource cost, and it is not comparable with audit-trail budgets.

Δ_K turn the comparison in favour of disclosure. For a known pair at coarse separation $\Delta\eta = 0.20$ ($\delta = 0.10$), weekly renewal of price-only probing costs \$0.32 million per market per year at Roll noise and \$7.79 million at one-second noise. At $\Delta\eta = 0.10$ ($\delta = 0.05$) the unknown $K = 2$ Gauss–Radau pair costs \$14.74 million (Roll) or \$356.44 million (one-second) per market per year; with $K = 3$ unknown cohorts at $\Delta\eta = 0.20$ the figures rise to \$41.83 million and \$1.01 billion, and at $K = 3$, $\Delta\eta = 0.10$ or $K = 4$, $\Delta\eta = 0.20$ they run to billions. For orientation only, CAT LLC's revised 2026 budget is \$147.0 million; as explained above, the two numbers measure different things. The renewal frequency is an assumption: annual figures scale with it, so monthly renewal divides them by 4.3 and renewal on each of 252 trading days multiplies them by 4.8.

S9 Account-level capital

Lemma 4.5 gives the order but not the constant. This section computes what a trader actually needs under a calibrated linear-impact environment, then converts it to Bitcoin dollars.

Numerical values. Decimal optima and brackets below are numerical evaluations of the stated bounds.

S9.1 Observation design and the tri-region partition

An observer distinguishes two candidate mechanisms with rates θ_0 and $\theta_1 = \theta_0 + \delta$ over N periods, submitting probes $V = (V_1, \ldots, V_N)^\top$ that generate inventory $I_t = \sum_{s \le t} V_s$. Under contract I_0 (active probing, Gaussian observation noise) $p_t = (h_i * V)_t + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$, so

$$D_{\mathrm{KL}}(P_0^{\otimes N} \| P_1^{\otimes N}) = \frac{1}{2\sigma^2} \sum_{t=1}^{N} \big((h_1 * V)_t - (h_0 * V)_t\big)^2. \tag{S2}$$

The observer minimizes capital $C(N, V) = \sum_t (\lambda V_t^2 + \phi |V_t|) + \max_t \mu p_0 |I_t|$ subject to (i) separation $D_{\mathrm{KL}} \ge \frac{1}{2}(z_\alpha + z_\beta)^2$, (ii) caps $|V_t| \le V_{\mathrm{eff}}$ and $|I_t| \le I_{\max}$, and (iii) a solvency budget $\max_t \mathrm{Loss}_t(V) \le B$. Here $V_{\mathrm{eff}} = \min(V_{\max}, D_{\max}/\kappa)$ combines resting depth and a price-displacement limit.

Proposition S9.1 (Tri-region partition and finite-session resolution floor)**.** *Under observation horizon T per pulse, (δ, B) space splits into three regions.*

(i) Feasible single pulse. *For $\delta \ge \delta_{\min}(T)$ and $B \ge C^*(\delta)$ one pulse within capacity suffices. For a simple rate shift ($\eta_1 = \eta_0 + \delta$) the response difference is $\Theta(\delta)$ and $C^* = \Theta(\delta^{-2})$; for symmetric splitting the first-order variation cancels, the difference is $\Theta(\delta^2)$ and $C^* = \Theta(\delta^{-4})$.*

(ii) Sequential bounded probing. *For $\delta < \delta_{\min}(T)$ the required quantity exceeds V_{eff}. Capped pulses each supply divergence $V_{\mathrm{eff}}^2 \|d_T\|^2/(2\sigma^2)$, so*

$$N_{\mathrm{pulses}}(T, \delta) = \Big\lceil \frac{4\sigma^2 z_\alpha^2}{V_{\mathrm{eff}}^2 \|d_T\|_2^2} \Big\rceil. \tag{S3}$$

(iii) Infeasible with isolated pulses. *If the budget cannot finance sequential costs, peak inventory breaches $I_{\max}$, or a session deadline binds, the finite-session floor $\delta_{\min}(T_{\mathrm{session}})$ is the root of $V_{\mathrm{eff}} \|d_{T_{\mathrm{cycle}}}(\delta)\|_2 = 2\sigma z_\alpha / \sqrt{\lfloor T_{\mathrm{session}}/T_{\mathrm{cycle}} \rfloor}$, with small-$\delta$ approximation $\delta_{\min} = \Theta(T_{\mathrm{session}}^{-1/4})$.*

In words. A single test trade works only when the groups are far enough apart and the account has enough capital. Closer groups need many capped test trades, and a longer observation window after each trade makes every test trade more informative.

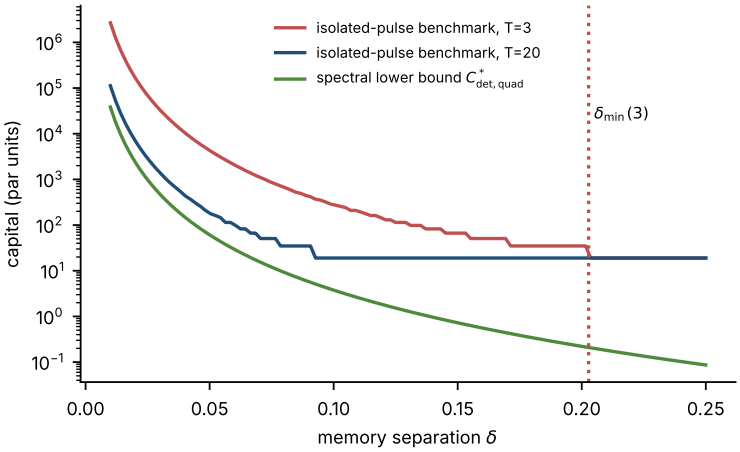


Figure S1: Capital against separation: isolated-pulse benchmarks with three and twenty observations per pulse, and the spectral lower bound of Theorem S9.2. The dotted line is the single-tap resolution floor $\delta_{\min}(3) = 0.2028$. Recomputed in this revision.

At the calibration $\kappa = 0.8$, $q = 0.2$, $\sigma = 0.01$, $V_{\max} = 50$, $D_{\max} = 5$ ($V_{\text{eff}} = 6.25$), the single-tap floor is $\delta_{\min}(3) = \sqrt{2\sigma z_\alpha/(V_{\text{eff}}\kappa^2 q)} = 0.2028$; extended observation lowers it to 0.1376 ($T = 4$), 0.0973 ($T = 10$), 0.0914 ($T = 20$) and 0.0905 ($T = 100$). At $\delta = 0.05$ the simplified one-leg benchmark (cost $\frac{1}{2}\kappa V_{\text{eff}}^2 + \phi P_* V_{\text{eff}} = 15.875$ per pulse plus initial margin 3.125) needs 271 pulses and 4,305.25 par units with $T = 3$, 58 pulses and 923.88 with $T = 4$, but only 12 pulses and 193.63 with $T = 20$ and 11 pulses and 177.75 with $T = 100$; optimizing the pulse size (6.06 rather than 6.25) lowers the $T = 20$ figure to 182.03. Most of the saving comes from observing the transient response rather than a single tap (Figure S1).

These are costs within an isolated-pulse design class, not minimum capital; the next subsection replaces the formula by a ledger.

S9.2 Account-level capital and the gap to the lower bound

We run each candidate design through an explicit synthetic ledger at reference price $P_* = 100$: fills at the average linear-book price, taker fees of 4 basis points, marks at the prevailing price, and 0.5% maintenance margin. Initial capital is $C_0 = \max_t(\text{margin}_t - \text{equity}_t)$, maximized over both hypotheses, and the test uses the full price path including unwind legs.

Table S7 gives three conclusions. First, twelve buys that are never unwound fail the 5% target once the test uses the joint price path (divergence $4.95 < 5.41$) and end with 75 units of inventory. Second, if every unwind leg is itself informative, five or six round trips meet the target with zero terminal inventory. Third, pulses are not optimal: a generalized-eigenvector probe, which maximizes divergence per unit of fee-free round-trip cost, needs about 65 par

Table S7: Account-level initial capital at $\delta = 0.05$ (par units)

Design	C_0	Error	End inv.	Peak margin	Max $\lvert v_t \rvert$	Trades
12 buys, $T = 20$ apart, never unwound	229.13	0.0578	75.00	39.38	6.25	12
6 buy/sell round trips	189.45	0.0308	0.00	3.28	6.25	12
5 buy/sell round trips	157.89	0.0442	0.00	3.28	6.25	10
12 buys, each unwound after 20 steps	378.82	0.0040	0.00	3.28	6.25	24
Generalized-eigenvector probe ($L = 120$)	65.06	0.0500	0.00	2.74	1.24	120
Generalized-eigenvector probe ($L = 160$)	64.86	0.0500	0.00	2.17	1.09	160
Fee-refined probe ($L = 240$)	64.48	0.0500	0.00	—	1.30	240

The pulse designs place a pulse every 40 steps and close it 20 steps later; the fee-refined probe is re-optimized by sequential quadratic programming over $L = 240$. The numerically evaluated lower bound of Theorem S9.2 is 61.41; the explicit constant of Lemma 4.5 is 4.9×10^{-4}.

units with maximum trade 1.25 and maximum displacement 2.23, well inside limits.

For flat fee-paying policies $v \in \mathbb{R}^L$ with $\sum_t v_t = 0$, the capital problem is

$$\min_{v} \max_{i\in\{0,1\}} L_i(v) + \sum_t \phi P_* |v_t| + \max_t \mu P_* |I_t| \quad \text{s.t.} \quad \|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}, \tag{S4}$$

where $L_i(v) = \frac{1}{2} v^\top T_{h_i} v$ is the exact Toeplitz round-trip loss. Zero-padding embeds horizon L into $L+1$ at identical cost and information, so the minimum is non-increasing in L.

Theorem S9.2 (Spectral dissipation lower bound)**.** *Let $\hat{H}_i(z) = \sum_{j\geq 0} h_{i,j} z^{-j}$, $\Delta\hat{H} = \hat{H}_1 - \hat{H}_0$, and define the dissipation kernel*

$$g_i(\omega) = \operatorname{Re} \hat{H}_i(e^{i\omega}) - \kappa/2 > 0 \qquad \forall \omega \in [-\pi, \pi]. \tag{S5}$$

(i) Toeplitz energy identity. *For any flat round trip, $L_i(v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g_i(\omega) |V(e^{i\omega})|^2 d\omega$.*

(ii) Lower bound. $\|(H_1 - H_0)v\|_2^2 \leq \sup_\omega \frac{|\Delta\hat{H}(e^{i\omega})|^2}{g_i(\omega)} L_i(v)$, *so every deterministic round trip with error at most α has*

$$L_i(v) \;\geq\; 4\sigma^2 z_{1-\alpha}^2 \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2} \;\equiv\; C^*_{\text{det,quad}}(\delta). \tag{S6}$$

(iii) Bracket. *At $\delta = 0.05$, $C^*_{\text{det,quad}} = 61.14$ at $\omega^* = 0.2526$; adding the unavoidable fee $\phi P_* \cdot 2\sigma z / \sup_\omega |\Delta\hat{H}|$ gives the numerical lower estimate $C_{\text{det}} \geq 61.41$. A feasible 240-step sequence attains 64.48, so the fee-paying optimum lies in* $[61.41, 64.48]$.

(iv) Process noise. *Let the price also receive the innovations $w_t \sim N(0, \sigma_w^2)$ of (1), independent of observation noise $\nu_t \sim N(0, \sigma_\nu^2)$, so that from the reset state the observer sees $\tilde{x}_t = (h_i * v)_t + \kappa^{-1}(h_i * w)_t + \nu_t$, $t < T$. For the known pair of Theorem 4.1,*

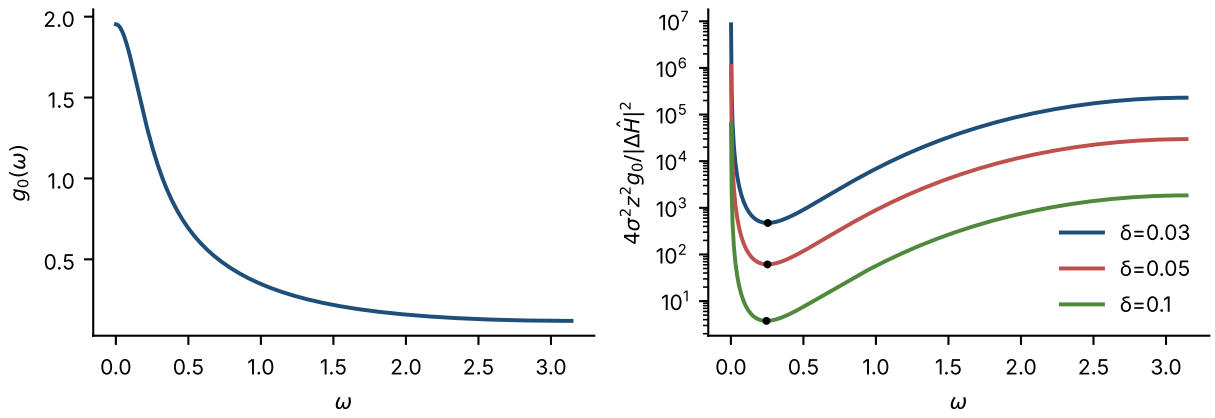


Figure S2: Left: the dissipation kernel $g_0(\omega)$ is strictly positive. Right: the capital ratio in (S6); dots mark its minimum, near $\omega^* \approx 0.25$ for every separation. Recomputed in this revision.

put $\sigma_{\text{eff}}^2 = \sigma_\nu^2 + \sigma_w^2/4$, $\bar{M} = 1/(1-S)$ *and* $C_w = 2\sigma_w^4 \bar{M}^2 K_*^2(\sigma_\nu^2 + \sigma_w^2 \bar{M}^2)/(\kappa^2 \sigma_\nu^2 \sigma_{\text{eff}}^4)$. *Every deterministic round trip whose test has errors* α *and* β *under the two hypotheses satisfies*

$$L_i(v) \;\geq\; 2\sigma_{\text{eff}}^2 \big[\mathrm{kl}(\alpha, 1-\beta) - C_w T \delta^4\big]_+ \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{S7}$$

with kl *as in Theorem S9.3. In-loop noise therefore leaves the* δ^{-4} *capital order intact for every horizon* $T \leq \mathrm{kl}(\alpha, 1-\beta)/(2C_w\delta^4)$*; at horizons of order* δ^{-4} *and longer, the noise itself separates the pair without a probe (Theorem 5.8).*

In words. This turns the order into a dollar floor. No round trip that tells the two candidate markets apart at 5% error can lose less than this amount, and a concrete probe comes within 5% of it.

Positivity of g_i follows from Lemma 2.3: $g_i(\omega) = \frac{1}{2}\sum_\ell \omega_\ell \frac{1-\lambda_\ell^2}{1-2\lambda_\ell \cos\omega + \lambda_\ell^2} > 0$. We compute $C^*_{\text{det,quad}}$ across separations (Table S8): 61.14 at $\delta = 0.05$, with $\delta^4 C^*_{\text{det,quad}}$ between 3.67×10^{-4} and 3.84×10^{-4}, confirming the δ^{-4} law. The fee term, the feasible upper bounds and the adaptive bound of Theorem S9.3 are reported in the same table.

The infimum over ω is evaluated on a fine frequency grid; a local refinement agrees to four digits (61.1377 at $\omega^* = 0.2526$), and the reported constant is the larger of the two hypotheses' bounds. "Certified" here means a numerical evaluation of the stated bound, not an interval-arithmetic enclosure.

Theorem S9.3 (Adaptive and sequential probing)**.** *Let* $(v_t)_{t\leq\tau}$ *be any non-anticipating policy adapted to the price filtration, with an almost surely finite stopping time and* $\sum_{t\leq\tau} v_t = 0$,

Table S8: Capital lower bounds and feasible upper bounds across separations (par units)

δ	ω^*	$C^*_{\mathrm{det,quad}}$	$\delta^4 C^*_{\mathrm{det,quad}}$	Certified C_{det}	Feasible UB	Adaptive LB
0.02	0.2546	2397.85	3.84×10^{-4}	2399.58	2428.25	1176.04
0.03	0.2541	473.19	3.83×10^{-4}	473.97	484.50	232.51
0.05	0.2526	61.14	3.82×10^{-4}	61.41	64.48	30.22
0.08	0.2488	9.26	3.79×10^{-4}	9.37	10.38	4.64
0.10	0.2453	3.77	3.77×10^{-4}	3.83	4.44	1.91
0.15	0.2331	0.73	3.67×10^{-4}	0.76	1.01	0.38

Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, $\sigma = 0.01$, $\alpha = 0.05$. the numerical-bound column adds the fee lower bound to the fee-free infimum; the feasible column is the best flat probe found over $L \leq 240$; the adaptive column adds the fee lower bound to the adaptive bound.

followed by a terminal test $\psi_\tau \in \{0,1\}$ *with* $P_0(\psi_\tau = 1) \leq \alpha$ *and* $P_1(\psi_\tau = 0) \leq \beta$. *Then*

$$\mathbb{E}_0[L_0(v)] \;\geq\; 2\sigma^2 \,\mathrm{kl}(\alpha, 1-\beta) \inf_{\omega} \frac{g_0(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{S8}$$

where $\mathrm{kl}(\alpha, 1-\beta) = \alpha \log \frac{\alpha}{1-\beta} + (1-\alpha) \log \frac{1-\alpha}{\beta}$. *Hence adaptation can lower expected fee-free loss by at most the factor* $\mathrm{kl}(\alpha, 1-\beta)/(2z^2_{1-\alpha}) \approx 0.4897$ *at* $\alpha = \beta = 0.05$*, and cannot alter the* δ^{-4} *order.*

In words. Letting the observer adapt its test trades as data arrive, and stop as soon as it is confident, helps only a little: at the calibration it can cut the expected loss by at most about half. The steep cost is a property of the market, not of a clumsy probing plan.

At $\delta = 0.05$ the adaptive bound is 29.94, or 30.22 with the minimal fee. Pairing it with the feasible deterministic 64.48 gives a numerical ratio of at least 0.464. In 20,000 simulated sequential probability-ratio tests, which split the optimal probe into ten copies of equal information, mean loss was 39.5 under H_0, between the two bounds, but losses vary with the stopping time: the 95th and 99th percentiles of required account capital were 91.7 and 128.4, and funding a trader with only the mean loss gives a 37.5% insolvency rate before the test ends. Expected-loss savings are not solvency savings.

Theorem S9.4 (Finite-horizon Rayleigh–Hölder duality)**.** *For a horizon of* $L \geq 4$ *trading periods, let* $E \in \mathbb{R}^{L \times (L-1)}$ *be the first-difference basis of flat probes (*$\sum_t v_t = 0$*),* $Q_i = \frac{1}{2} T_{h_i}$ *(so that* $L_i(v) = v^\top Q_i v$*),* $A_{\mathrm{diff}} = (H_1 - H_0)^\top (H_1 - H_0)$*, and let* $G_i = E^\top Q_i E \succ 0$ *and* $M = E^\top A_{\mathrm{diff}} E \succeq 0$ *be the reduced execution-loss and information matrices, with* G_i*-orthonormal generalized eigenvectors* e_j *and eigenvalues* $0 \leq \lambda_1 \leq \cdots \leq \lambda_{L-3} < \lambda_{L-2} \leq \lambda_{L-1}$*; put* $c_{\mathrm{req}} = 4\sigma^2 z^2_{1-\alpha}$ *and* $c_{\mathrm{fee}} = \phi P_*$*. For* $\theta \in [0, \pi)$ *let* $d(\theta) = \cos\theta \, e_{L-2} + \sin\theta \, e_{L-1}$*,* $\lambda(\theta) =$

$\lambda_{L-2}\cos^2\theta + \lambda_{L-1}\sin^2\theta$, $s_\theta = \operatorname{sign}(Ed(\theta))$, *and*

$$F_{\mathrm{dom}}(\theta) = c_{\mathrm{fee}}\sqrt{\frac{c_{\mathrm{req}}}{\lambda(\theta)}}\,\|Ed(\theta)\|_1, \qquad W_\perp(\theta) = c_{\mathrm{fee}}^2 \sum_{j\le L-3} \frac{(s_\theta^\top E e_j)^2}{1-\lambda_j/\lambda(\theta)}, \tag{S9}$$
$$\gamma(\theta) = \frac{\lambda(\theta)\,\lambda_{L-3}}{(\lambda(\theta)-\lambda_{L-3})\,c_{\mathrm{req}}}.$$

Then every flat probe meeting the separation constraint of (S4) *satisfies*

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\ge\; \min_\theta\Big\{\frac{c_{\mathrm{req}}}{\lambda(\theta)} + \Psi(\theta)\Big\} \;\equiv\; C_{\mathrm{RH}}(L) \;\ge\; \frac{c_{\mathrm{req}}}{\lambda_{L-1}}, \tag{S10}$$

$$\Psi(\theta) = \min_{x\ge 0}\Big\{x^2 + \Big[F_{\mathrm{dom}}(\theta)\big(1-\gamma(\theta)x^2\big)_+ - \sqrt{W_\perp(\theta)}\,x\Big]_+\Big\} \;\ge 0,$$

a piecewise-quadratic minimization in closed form; when $\gamma F_{\mathrm{dom}} < 1$ *and the minimizer has* $\gamma x^2 \le 1$, $\Psi \ge \max\big(0, F_{\mathrm{dom}} - W_\perp/(4(1-\gamma F_{\mathrm{dom}}))\big)$, *which is what the bound uses.*

In words. This is the exact cheapest probe over a finite number of trading periods, fees included. It confirms that the dollar floor of Theorem S9.2 is not an artefact of assuming an infinitely long probe.

The bound is computable for any horizon. Table S9 reports its quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$, which decays toward the infinite-horizon value 61.14 at rate $O(L^{-1})$, the full bound $C_{\mathrm{RH}}(L)$, and the best feasible fee-paying probe we find at each horizon. At $L = 80$ the full bound is 64.72 against a best probe of 64.76 on the same loss-plus-fee objective (Table S9 reports 65.02 for ledger capital, which also includes margin), so the dual is essentially tight at short horizons; at longer horizons the fee term of the dual weakens and the bracket widens to about 5%. The fee of unpadded full-support eigenvectors grows as $\Theta(\sqrt{L})$, which is why zero-padded or L_1-localized probes are needed at long horizons.

S9.3 Robustness: heteroskedastic impact and probing protocols

Corollary S9.5 (Invariance under impact-induced heteroskedasticity)**.** *Suppose probing inflates innovation variance as* $\sigma_{i,t}^2(v) = \sigma^2(1+\xi[(h_i * v)_t]^2)$ *and slippage scales as* $\kappa_{\mathrm{eff}} = \kappa(1+\xi_0|v_t|)$. *For capacity-bounded probes,* $D_{\mathrm{KL}}(P_0^v\|P_1^v) \le \frac{1+O(\xi^2 V_{\mathrm{eff}}^2)}{2\sigma^2}\|(H_1-H_0)v\|_2^2 = O(\|v\|_2^2\delta^4)$, *and convex impact only raises dissipation. Required capital remains* $\Theta(\delta^{-4})$.

Four probing protocols (unconstrained single pulse, repeated capped pulses, a sequential test with dynamic throttling, and a predetermined generalized-eigenvector trajectory) all show capital slopes near -4 in $\log\delta$: -3.90, -3.84, -3.47 and -3.81 over $\delta \in [0.03, 0.12]$,

Table S9: Finite-horizon capital at $\delta = 0.05$ (par units)

Horizon L	Quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$	Full bound $C_{\mathrm{RH}}(L)$	Best feasible probe	Gap to $C_{\mathrm{det}} = 61.41$
80	62.55	64.72	65.02	5.9%
120	61.84	63.39	64.59	5.2%
160	61.54	62.46	64.50	5.0%
240	61.32	61.76	64.48	5.0%
320	61.25	—	≤ 64.48	$\leq 5.0\%$
480	61.19	—	≤ 64.48	$\leq 5.0\%$

Quadratic bound $c_{\mathrm{req}}/\lambda_{L-1}$ from the generalized eigenproblem of Theorem S9.4 on flat probes (equivalently $c_{\mathrm{req}}/(2\lambda'_L)$ with λ' computed against T_{h_0} rather than $\frac{1}{2}T_{h_0}$); it converges to the infinite-horizon value 61.14. The full bound minimizes (S10) over 3,600 values of θ; at $L = 80$ and 120 a direct numerical minimization of loss plus fee under the separation constraint gives 64.76 and 64.43, above the bound. Feasible probes: top generalized eigenvector refined by sequential quadratic programming on the full ledger; at $L = 320$ and 480 the $L = 240$ probe padded with zeros is feasible at the same cost, so the best feasible value can only be lower.

Table S10: Capital lower bound under alternative observation-noise calibrations (USDT)

Noise calibration	$\delta = 0.02$	$\delta = 0.05$	$\delta = 0.10$
Roll estimator, median (0.40 USDT)	3,874,696	99,192	6,200
One-second price-change s.d., median (1.98 USDT)	93,688,369	2,398,422	149,901
Earlier assumption (4 USDT, upper scenario)	383,812,500	9,825,600	614,100

$C(\sigma, \delta) = 61.41 \times (\sigma/0.01)^2 \times (0.05/\delta)^4$, with σ the unrounded median. Noise estimated on 11 days of checksum-verified Binance BTCUSDT one-second klines.

where the sequential slope is flattened by the 1.7 pulses needed at $\delta = 0.12$ (Appendix Table S2). At equal observation horizon, sequential stopping saves 14–49% of capital relative to repeated capped pulses, but most of the apparent saving from the single-tap baseline comes from observing the transient response, not from adaptivity.

S9.4 From par units to Bitcoin dollars

Capital scales as $\sigma^2\delta^{-4}$, so the dollar magnitude hinges on the observation noise. Treating the 4 USDT ten-second price volatility as observation noise would give 9.8 million USDT at $\delta = 0.05$, but that conflates price volatility with measurement error. We now estimate observation noise from Binance BTCUSDT one-second closes on January 10–20, 2023, with the Roll covariance estimator $\hat{\sigma} = \sqrt{-\mathrm{cov}(\Delta p_t, \Delta p_{t-1})}$ (Roll, 1984). Daily estimates range from 0.09 to 0.77 USDT, median 0.40 USDT.

At realistic noise, separating rates that differ by 0.05 costs about 9.9×10^4 USDT, which is not prohibitive for a well-capitalized desk (Table S10). The barrier becomes large only for finer separation (3.9 million USDT at $\delta = 0.02$) and, through Theorem 5.3, when several cohorts are unknown. Two factors we do not model would raise these figures: walking

Table S11: Daily observation-noise estimates, Binance BTCUSDT one-second closes

Date	Roll $\hat{\sigma}$ (USDT)	One-second change s.d. (USDT)	Mean price (USDT)
2023-01-10	0.152	0.679	17,288
2023-01-11	0.092	0.891	17,456
2023-01-12	0.232	2.292	18,343
2023-01-13	0.348	1.762	19,075
2023-01-14	0.773	3.511	20,818
2023-01-15	0.482	1.790	20,778
2023-01-16	0.629	2.357	21,036
2023-01-17	0.457	1.976	21,194
2023-01-18	0.695	2.541	21,122
2023-01-19	0.402	1.316	20,850
2023-01-20	0.316	2.038	21,306
Median	0.402	1.976	

The Roll estimator measures the negative first-order autocovariance of price changes, which reflects bid-ask bounce and recording noise rather than price volatility.

several levels of a real book costs more than the linear top-of-book impact assumed here, and impact-induced heteroskedasticity raises the noise a probe must overcome. The capital figures are therefore lower bounds for the stated model.

S10 Near-colliding rates: a uniform rate for the merged measure

Theorem 5.4 needs the rates to stay γ-separated and every weight to stay away from zero. Neither restriction can be dropped for weight recovery: two atoms of weights w and $W - w$ at rates r and $r+\Delta$ and two of weights $W-w$ and w at the same rates have merged measures that converge to the same measure as $\Delta \to 0$, so no estimator recovers the individual weights uniformly over near-colliding economies. What can be recovered uniformly is the merged measure itself, in the permutation-invariant Wasserstein distance.

Theorem S10.1 (Uniform rate without separation)**.** *Fix $K \geq 1$, a memory bound $\bar{r} < 1$ and $0 < m_{\min} \leq B$. Let $\mathcal{M}_K$ be the merged measures with at most K atoms in $[0, \bar{r}]$ and total mass $k_0 \in [m_{\min}, B]$, with no separation condition and no lower bound on individual weights, and write $\bar{M} = M/k_0$ for the normalized memory distribution. From N reset experiments as in Theorem 5.4, let $\hat{M}$ be any element of $\mathcal{M}_K$ whose first $2K$ kernel values are closest in the sup norm to the estimated ones. There is C depending only on $(K, \bar{r}, a, \kappa, B)$ such that,*

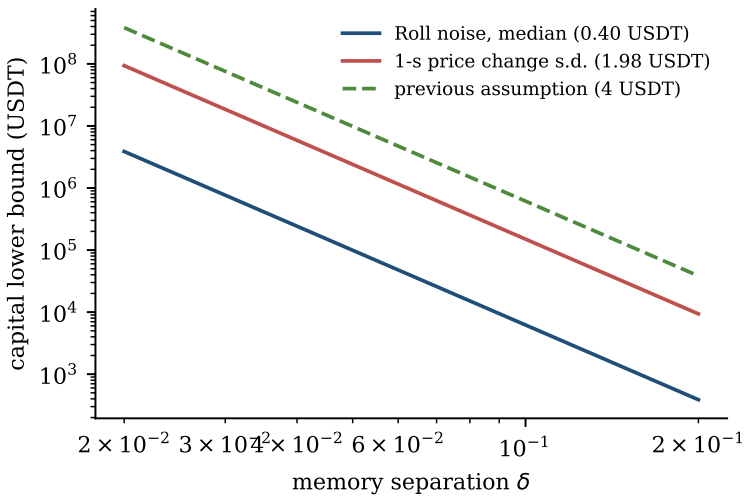


Figure S3: Capital lower bound against memory separation for three observation-noise calibrations.

with probability at least $1-\beta$,

$$W_1(\hat{\bar{M}}, \bar{M}) \;\le\; C\Big(\frac{\sigma\sqrt{\log(4K/\beta)}}{m_{\min}\sqrt{N}}\Big)^{1/(2K-1)} \qquad \textit{uniformly over } \mathcal{M}_K.$$

Hence $N = O(\sigma^2 m_{\min}^{-2}\epsilon^{-(4K-2)})$ *experiments give accuracy* ϵ*, and with the Wasserstein statement of Theorem 5.3 the minimax rate for the merged memory distribution is* $N^{-1/(4K-2)}$*, uniformly over near-colliding economies.*

In words. Even with no lower limit on how close the groups' memories are or how small a group is, the combined memory distribution can still be estimated, and no faster rate is possible. What is lost when groups nearly coincide is the ability to tell them apart, not the ability to describe the combined picture.

Proof. As in the proof of Theorem 5.4, the kernel values $k_0, \dots, k_{2K-1}$ are a Lipschitz function of the first $2K+1$ responses on the bounded set of responses the class produces, so a union bound gives $\max_{t<2K} |\bar{k}_t - k_t| \le e = C_1\sigma\sqrt{\log(4K/\beta)/N}$ with probability at least $1-\beta$ (the Lipschitz constant is for a bounded neighbourhood of the class; when the noise is too large for that, the right-hand side of the theorem exceeds $\bar{r} \ge W_1$ once C is enlarged, so the bound holds trivially). The class is compact and the moment map continuous, so a closest element $\hat{M}$ exists, and $\max_t |k_t(\hat{M}) - k_t(M)| \le 2e$ by the triangle inequality. Since all atoms lie in $[0,1]$, $|k_t| \le k_0$, and the normalized moments $m_t = k_t/k_0$ satisfy $|m_t(\hat{M}) - m_t(M)| \le 2e/k_0 + (|\hat{k}_t|/\hat{k}_0)(2e/k_0) \le 4e/m_{\min}$. The affine map $r \mapsto 2r/\bar{r} - 1$ carries

both normalized measures to distributions on $[-1, 1]$ with at most K atoms, multiplies W_1 by $2/\bar{r}$ and the differences of moments of order below $2K$ by at most $(1 + 2/\bar{r})^{2K-1}$. The moment comparison inequality for K-atomic distributions on $[-1, 1]$ (Wu and Yang, 2020), $W_1 \le c_K \delta^{1/(2K-1)}$ when the first $2K - 1$ moments differ by at most δ, then gives the bound with $C = c_K(\bar{r}/2)(1 + 2/\bar{r})(4C_1)^{1/(2K-1)}$. □

The theorem separates two questions that Theorem 5.4 answers together. Which memory distribution do the participants have? That has the same polynomial cost whether or not their rates collide. Which participant has which weight? That becomes arbitrarily expensive as rates approach each other (Theorem 5.3) and impossible when they coincide (Theorem 3.1). A supervisor who needs the first can have it at the rate of Theorem 5.3; one who needs the second has to be told the number of cohorts and a separation, or has to obtain tagged data (Section 6).

S11 Robustness to nonlinear and adaptive trading rules

The theorems of Sections 3–5 use the linear exponential-memory recursion (2). Real desks cap positions, trade only beyond thresholds, adapt their gains and learn online. A small probe of size A moves the state by $O(A)$, so any rule whose order is a twice-differentiable function of an exponential-memory state is linear to first order and differs from it by $O(A^2)$; the theorems should then hold for small probes and fail for large ones. We check this, and locate where it fails, in simulations of five stylized rules at $a = 0.5$, $\kappa = 0.8$, total weight $q = 0.2$ and mean rate $\bar{\eta} = 0.4$, over 240 periods after a pulse probe.

The rules are: (1) *momentum with a position limit*, the cohort's order a smooth saturating (tanh) function of its memory state $y_{i,t}$, normalized to unit slope at the operating point; (2) a *threshold rule* that trades only when $|y_{i,t}| > \tau = 0.05$, in two versions, an exact dead zone and one averaged over background state variation of standard deviation 0.04; (3) an *adaptive-gain rule* in the form of recursive least squares with forgetting factor $1 - \eta_i$, whose gain falls when the regressor is large; (4) an *online neural readout*, a linear readout of a fixed two-layer tanh feature map with eight hidden units, trained by regularized stochastic gradient steps of size η_i; and (5) a *tabular value rule* that updates a 21-bin table at rate η_i, in two versions, kernel-smoothed across bins and a single-bin lookup. These are stylized: each is built so that its small-signal dynamics are an exponential filter with rate η_i, and they are not calibrated to any production system.

Two checks follow the theory. *Test A (merged measure):* one cohort with weight q and

position limit 1 against two cohorts with weights $q/2$ at the same rate and limits 0.55 and 1.65, so the two economies have the same merged measure; the raw response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2$ should be $O(A^2)$ (slope 2 in $\log A$) and the probe-normalised gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ should be $O(A)$ (slope 1 in $\log A$) over $A \in [0.002, 0.064]$. *Test B (δ^{-4} law):* one cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$, $\delta \in [0.015, 0.12]$, at probe size $A = 0.005$; the normalized response difference $\|\Delta h\|_2$ should scale as δ^2 and the capital lower bound of Theorem S9.2 as δ^{-4}. The saturation threshold A_{sat} is the smallest probe at which $\|\Delta h\|_2$ at $\delta = 0.05$ departs by more than 10% from its small-probe value.

To check that these conclusions do not depend on hand-built rules, we pre-specified a second experiment in which online learners are trained from scratch inside the closed-loop market for $T_{\text{burn}} = 5{,}000$ periods of flow and noise innovations; their learned parameters are saved and reloaded into the probing simulator without rescaling. Four learning rules are used: (i) recursive least squares with forgetting factor $\lambda = 0.995$, learning its return-forecast slope; (ii) a two-layer tanh network with eight hidden units whose readout is trained by regularized online stochastic gradient descent on trading profit; (iii) a scalar policy-gradient rule trained on trading profit; and (iv) a tabular Q-learner with 21 states and 21 actions, TD(0) Bellman backups ($\gamma = 0.90$) and Boltzmann action choice. Each learner acts on an exponential-memory state $y_{i,t}$ with rate η_i: training sets its response gain and curvature, not its memory rate, so the experiment asks whether learned policies preserve the merged-measure and δ^{-4} predictions, not whether learning produces exponential memory. The specification fixed four predictions: (a) a single exponential fits each trained learner's small-probe impulse response with $R^2 \geq 0.90$; (b) two trained populations with the same merged measure differ by at most 1% in relative L_2 response at $A = 0.005$; (c) the capital lower bound to separate trained populations whose memory rates differ by 2δ scales with log–log slope in $[-4.15, -3.85]$; and (d) we report where equivalence and the δ^{-4} law fail when learning continues during the probe.

Table S12 and Figure S4 show both where the linear reduction holds and where it breaks down. In Panel A, for small probes every smooth or dithered rule behaves like the linear model: economies with the same merged measure have raw response gap $O(A^2)$ (slopes 2.011 to 2.064) and probe-normalised gap $O(A)$ (slopes 1.011 to 1.064), while the response difference and capital lower bound scale as δ^2 (slope 2.007) and δ^{-4} (slope -4.001), with capital at $\delta = 0.05$ within 0.1% of the linear 61.42. In Panel B, predictions (a)–(c) pass for all four trained learners: a single exponential explains $R^2 \geq 0.993$ of each small-probe impulse response, populations with the same merged measure differ by at most 10^{-5} in relative response at $A = 0.005$, and capital lower bound scales with slopes -4.014 to -3.973, inside the registered band. The level of capital differs across learners because training sets

Table S12: Merged-measure equivalence and the δ^{-4} law under stylized nonlinear rules and closed-loop trained online learners

Rule	Test A raw ($O(A^2)$)	Test A norm. ($O(A)$)	Test B slope ($\|\Delta h\|_2$)	Test B slope (capital)	Capital at $\delta = 0.05$	A_{sat} / R^2_{exp}
Panel A. Stylized nonlinear and adaptive rules (fixed operating point)						
Linear benchmark	—	—	2.007	−4.001	61.42	—
1. Momentum (tanh limit)	2.015	1.015	2.007	−4.001	61.47	0.834
2. Threshold rule, averaged	2.043	1.043	2.007	−4.001	61.42	0.049
3. Adaptive gain (RLS)	2.064	1.064	2.007	−4.001	61.46	0.751
4. Online neural readout	2.011	1.011	2.007	−4.001	61.48	0.751
5. Tabular value, smoothed	2.039	1.039	2.007	−4.001	61.44	1.271
Panel B. Learners trained in the closed loop, saved and reloaded (frozen during the probe; shared training stream)						
6. Recursive least squares	3.000	2.000	2.008	−4.014	54.58	$R^2 = 0.993$
7. Two-layer network (SGD)	1.878	0.878	2.007	−4.012	65.19	$R^2 = 0.994$
8. Policy gradient on profit	3.000	2.000	2.004	−4.006	58.05	$R^2 = 0.995$
9. Tabular Q-learner (TD(0))	2.108	1.108	1.994	−3.973	186,821	$R^2 = 1.000$
Panel C. Where the trained-learner results change (range over the four learners)						
Independent training streams	rel. gap $\leq 10^{-5}$		0.13 to 0.92	−1.75 to −0.38	—	(c) fails
Learning during the probe	rel. gap $\leq 1.1 \times 10^{-3}$		1.99 to 2.01	−4.02 to −3.98	—	(c) holds
Exact dead zone ($\tau = 0.05$): $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe moves the state past τ only for $A > 0.139$.						
Single-bin value lookup: $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe leaves the central bin only for $A > 0.149$.						

Panel A: deterministic simulations around the operating point; Test A reports the raw response-gap slope in $\log A$ ($O(A^2)$) and the probe-normalised gap slope $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ ($O(A)$, plotted in Figure S4). Panel B: learners trained for 5,000 periods, parameters saved and reloaded without rescaling and frozen during the probe; the two compared populations are trained on the same innovation stream; relative equivalence gaps at $A = 0.005$ are at most 10^{-5}. A raw Test A slope above 2 means the gap is of higher order than A^2. Panel C: the same learners trained on independent innovation streams (first row) or updated at learning rate 0.35 during the probe (second row). Capital in par units, lower bound of Theorem S9.2 (61.42 for the linear model); the Q-learner settles on a small response gain (0.022), so separating it costs far more capital.

the response gain.

Panel C shows where this changes. When the two populations are trained on independent innovation streams, prediction (c) fails: the capital slopes are −1.75, −0.97, −1.62 and −0.38. Independently trained populations end with learned gains that differ by sampling error of order $T_{\text{burn}}^{-1/2}$, so they differ in merged weight as well as in memory rate, and a difference in weight is cheaper to detect than an $O(\delta^2)$ difference in rates. The δ^{-4} law therefore prices the separation of memory rates between populations whose feedback weights match; it is not a claim about telling apart any two trained populations. When learning continues during the probe at rate 0.35, the law survives (slopes −4.02 to −3.98) and the merged-measure gap rises only to 1.1×10^{-3}, so the breakdown we anticipated under prediction (d) did not occur in this design.

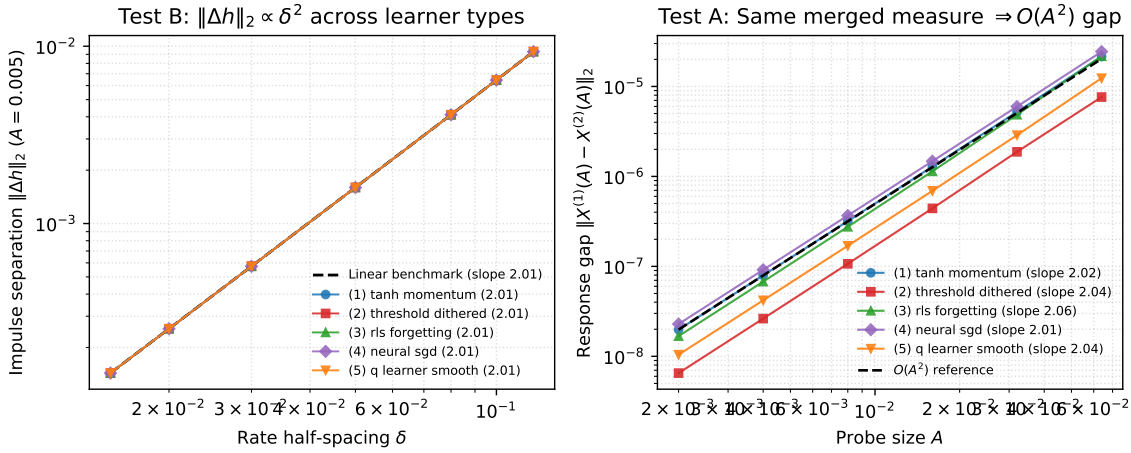


Figure S4: Left: Test B, probe-normalised response difference $\|\Delta h\|_2$ against the rate half-spacing δ at probe size $A = 0.005$, five stylized rules and the linear benchmark. Right: Test A, probe-normalised response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ between two economies with the same merged measure against the probe size A, with an $O(A)$ reference line (corresponding to an $O(A^2)$ raw gap).

The linear theory fails in two further places. First, saturation, adaptive gains and the neural readout leave the small-signal regime once probes exceed $A_{\text{sat}} \in [0.751, 1.271]$ par units, and the dithered threshold rule leaves it already at $A_{\text{sat}} = 0.049$, the threshold itself. Second, rules with hard discontinuities are different in kind: an exact dead zone ($\tau = 0.05$) or a single-bin lookup does not respond at all to probes smaller than 0.139 and 0.149 par units ($\|\Delta h\|_2 = 0$), concealing those cohorts from small probes altogether. Neither smooth nonlinearity nor hard thresholds offers a cheaper route to separating memory rates than the linear benchmark; populations whose learned weights differ are told apart by their weights, not their rates.

S12 A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

Theorem S12.1 (Spectral blindness)**.** *For any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance $s_1 > 0$ and stationary price variance V large enough, there is a family of two-state markets $J_\theta = \left(\begin{smallmatrix} \lambda_1 & \theta \\ 0 & \lambda_2 \end{smallmatrix}\right)$ with shock covariance $\operatorname{diag}(s_1, s_2(\theta))$ such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain $\sup_k \|e_1^\top J_\theta^k\|$ grows without bound in θ.*

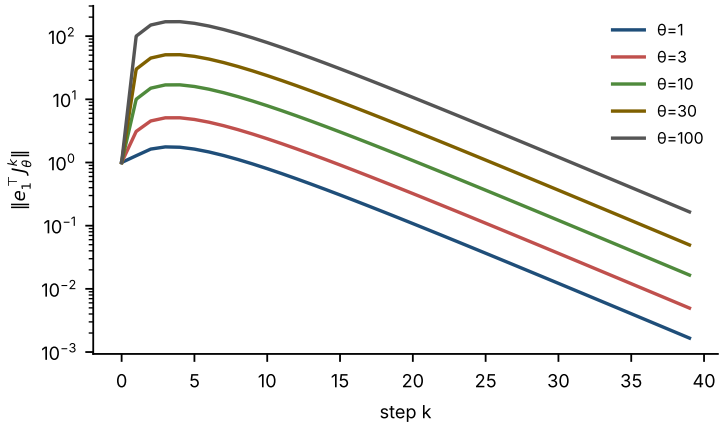


Figure S5: Price-row transient gain of the spectral-blindness family J_θ: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every θ.

In words. Two markets can look identical by the usual risk measures, the same eigenvalues and the same price variance, and yet one of them can produce far larger losses in the short run. Monitoring eigenvalues or volatility alone therefore cannot certify that a market is safe.

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure S5).

Two clarifications prevent over-reading. First, the cohort economy of Section 2 is itself non-normal, yet its price output does not amplify (Lemma 2.3). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (Supplement S3.5). Second, in a network of M identical cohort economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of Lemma 2.3; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the Kreiss constant stays at one, as it must for a stable symmetric operator. A stable normal contraction cannot exhibit Euclidean state-norm transient growth. This does not exclude scalar input-output overshoot relative to initial impact, which also depends on the input and output maps. The positive-cohort no-overshoot result is model-specific; the companion paper studies a depth-depletion mechanism.

S13 Empirical feedback and its measurement

The theory says what prices can reveal. The empirical work has two roles, and neither is a test of the identification bounds themselves, which concern controlled experiments no outside researcher can run. First, Section 6.4 tests the theory's policy prediction directly: that class-tagged data pin down memories that prices leave unrestricted. Second, this section asks what the reduced-form feedback indices a supervisor would compute from public data actually measure, and how much of them is feedback rather than filtering, timing or trading activity.

S13.1 Estimand and filtering mechanics

In theory $x_t = p_t - p_t^*$ is the deviation from an unobserved fundamental. Empirical work replaces p_t^* by a trailing filter. Let p_b be a log price proxy on uniform bars of length Δ and $m_b = (1-\alpha)m_{b-1} + \alpha p_b$ an exponential moving average with $\alpha = 2/(W/\Delta + 1)$. The empirical state is

$$x_b^{\mathrm{EMA}} \equiv p_b - m_b = (1-\alpha)(p_b - m_{b-1}) = \frac{1-\alpha}{1-(1-\alpha)L}\Delta p_b. \tag{S11}$$

Because $x_{b+1}^{\mathrm{EMA}} = (1-\alpha)(p_{b+1} - m_b)$, a martingale price gives

$$\mathbb{E}[x_{b+1}^{\mathrm{EMA}} \mid \mathcal{F}_b] = (1-\alpha)x_b^{\mathrm{EMA}} : \tag{S12}$$

the filtered state is an AR(1) with mechanical persistence $a_0 = 1-\alpha$, which equals $29/31 \approx 0.9355$ for $W = 300$ seconds and $\Delta = 10$ seconds.

Let f_t be signed flow. The empirical VAR(1)

$$\begin{pmatrix} x_{t+1}^{\mathrm{EMA}} \\ f_{t+1} \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \begin{pmatrix} x_t^{\mathrm{EMA}} \\ f_t \end{pmatrix} + e_{t+1} \tag{S13}$$

has reduced-form coefficients, not authenticated impact or learner parameters. When denominators are nonzero we use the scale-invariant indices

$$\hat{Q} = \frac{\hat{J}_{12}\hat{J}_{21}}{1-\hat{J}_{22}}, \qquad \hat{S} = \hat{J}_{11} + \hat{Q}, \qquad \hat{L} = \frac{\hat{Q}}{1-\hat{J}_{11}}. \tag{S14}$$

Equation (S12) implies that $\hat{J}_{11}$ must be compared with $1-\alpha$, not with zero: persistence that a naive reading attributes to feedback can be pure filtering.

Table S13: Samples

Sample	Coverage	Size	Source
Binance spot panel, 8 pairs	96 dates, 2021–2024	66.3M one-second bars	Binance archive
Binance spot fee-event windows, 8 pairs	±20 days around 2022-07-08 and 2023-03-22	640 pair-days	Binance archive
Binance USD-M perpetuals, 30 symbols	90 dates, 2023–2026	2,565 symbol-days	Binance archive
Binance BTCUSDT one-second panel	72 dates, 2021–2023	6.2M one-second bars	Binance archive
Binance BTCUSDT noise-calibration days	January 10–20, 2023	11 days	Binance archive

Binance files are verified against the published SHA-256 checksums (every file used). Sampled dates are the 10th and 24th of each month.

Table S14: Multi-year Binance spot panel: reduced-form indices under three timing constructions

Construction	$\hat{Q}$	Date interval	$\hat{S}$	Date interval	$\hat{Q} > 0$
Forward-paired flow (look-ahead)	0.0081424	[0.0072965, 0.0089882]	0.92074	[0.91665, 0.92482]	87.4%
Observed-flow EMA	0.0003733	[0.0002351, 0.0005116]	0.92072	[0.91640, 0.92504]	62.4%
Observed, 30 flow lags	0.0002580	[0.0001415, 0.0003745]	0.91738	[0.91167, 0.92308]	59.1%
Observed raw return	−0.0001555	[−0.0003732, 0.0000622]	−0.04051	[−0.05078, −0.03025]	49.2%
Placebo random walk	0.0000016	[−0.0000020, 0.0000052]	0.93510	[0.93485, 0.93534]	51.0%

768 study units (96 dates × 8 pairs), 6,634,752 ten-second transitions; equal-weight asset means within date, then date means with Student intervals. Mechanical persistence $29/31 = 0.93548$.

S13.2 Data

S13.3 Clock alignment dominates reduced-form feedback

We estimate (S13) on ten-second bars built from one-second Binance klines across a 96-date, eight-pair spot panel (the 10th and 24th of each month, 2021–2024) under three constructions (Table S14), and run the identical pipeline on a placebo random walk with each pair-day's volatility. Pairing the state with *next-bar* flow (a look-ahead) gives $\hat{Q} = 0.0081$; aligning flow with the bar in which it is observed lowers this by more than an order of magnitude, to 0.00037; and replacing the EMA state by raw returns gives an interval that includes zero. Mean $\hat{J}_{11}$ under observed flow is 0.9203, below the mechanical 0.9355, and the placebo reproduces the mechanical value (0.9351) with $\hat{Q}$ indistinguishable from zero.

Table S15: Asset means under forward-paired and observed flow

Pair	Forward-paired $\hat{Q}$	Observed $\hat{Q}$	$\hat{Q}_{30}$	Observed $\hat{S}$
BTC/USDT	0.0141870	0.0006073	0.0004891	0.93464
ETH/USDT	0.0130885	0.0005675	0.0002549	0.93511
SOL/USDT	0.0087763	0.0002129	0.0001100	0.91966
ADA/USDT	0.0067465	0.0003928	0.0003546	0.91452
DOGE/USDT	0.0058956	0.0002464	0.0001432	0.92496
BNB/USDT	0.0057420	0.0004479	0.0002762	0.89533
XRP/USDT	0.0057034	0.0002249	0.0001508	0.92362
LTC/USDT	0.0049997	0.0002870	0.0002853	0.91792

96 dates per pair. $\hat{Q}_{30}$ controls 30 ten-second bars of lagged flow. Placebo $|\hat{Q}| \leq 1.1 \times 10^{-5}$ for every pair.

Table S16: Feedback beyond filtering mechanics (Binance BTCUSDT, 72 days, 2021–2023)

	Real prices	Placebo (independent random walk)
Mean loop gain $\hat{Q}$	0.0045	0.0000009
95% CI (day bootstrap)	[0.0034, 0.0056]	[−0.0000006, 0.0000023]
Mean price persistence $\hat{J}_{11}$	0.9364	0.9357
Mechanical value $1 - 2/31$		0.9355

Placebo 95% band across days: $[-0.0000095, 0.0000156]$. This construction pairs the state with next-second flow; it is not comparable in level with the observed-flow ten-second index of Table S14.

S13.4 Feedback beyond filtering mechanics

We re-estimate feedback on a separate sample and add a placebo. We use Binance BTCUSDT one-second klines on 72 sampled days in 2021–2023. The state x_t is the log price deviation from its EMA (span 30 one-second bins), f_t is next-second signed taker flow, and a VAR(1) gives $\hat{Q}$. Because the filter alone produces persistence, we run the identical pipeline on a placebo: a random walk with the same one-second volatility, independent of the real flow.

Table S16 gives two findings. Feedback, in the sense of a loop gain the filter cannot produce, is detectable: the real $\hat{Q}$ lies far outside the placebo band. It is also small, and the price persistence that a naive reading would attribute to feedback is almost entirely mechanical (0.9364 against 0.9355). Because this construction pairs the state with next-second flow, part of $\hat{Q}$ reflects the timing effect of Table S14; the placebo shows only that the filter does not produce it.

S13.5 Order splitting and the estimation of feedback

Persistent order flow makes lagged-flow regressions inconsistent. Suppose flow follows $q_t = \gamma y_t + u_t$ with memory $y_t = (1 - \eta)y_{t-1} + \eta \Delta p_{t-1}$, order splitting $u_t = \rho u_{t-1} + \nu_t$, and

Table S17: Feedback estimators under order splitting (simulation, 500 replications)

Estimator	Coverage at $\gamma = 0.2$	Rejection at $\gamma = 0$	Bias	RMSE
OLS	0.000	1.000	0.892	0.893
Cochrane–Orcutt, true ρ	0.944	0.054	0.001	0.039
Cochrane–Orcutt, estimated ρ	0.916	0.080	0.002	0.043
2SLS	0.938	0.044	−0.002	0.116
LIML	0.938	0.044	−0.003	0.117

Newey–West standard errors (10 lags), $T = 3{,}000$. Median first-stage $F = 224$; a placebo instrument (future z) has $F = 0.62$. Estimators outside 93–97% coverage give invalid inference.

$\Delta p_t = \kappa q_t + \lambda z_t + e_t$ with an observed exogenous price shock z_t. Lagged returns are correlated with u_{t-1}, so OLS of flow on returns is biased even when $\gamma = 0$. Quasi-differencing ($q_t - \rho q_{t-1} = \gamma(y_t - \rho y_{t-1}) + \nu_t$) or instruments that move y through prices but are independent of u ($z_{t-1}, z_{t-2}, z_{t-3}$) restore consistency. We compare estimators over 500 replications with $\eta = 0.3$ and $\rho = 0.8$ (Table S17).

OLS is badly biased. Cochrane–Orcutt with the true ρ, 2SLS and LIML have correct size and coverage; Cochrane–Orcutt with an estimated ρ under-covers (Cochrane and Orcutt, 1949; Staiger and Stock, 1997). These are properties of estimators in a simulated market. On real data, a 2024 sub-panel of 192 spot units shows why the correction matters little at our bar length: with one VAR lag, the flow-equation residuals have essentially no first-order serial correlation (median $\hat{\rho} = -0.0002$), and Cochrane–Orcutt quasi-differencing moves the mean loop gain from 3.15×10^{-4} to 3.21×10^{-4} (post hoc). Order splitting shows up instead at longer lags, which the 30-lag control of Table S15 addresses. A real-data instrumental-variables estimate needs an instrument whose exclusion can be defended, which we do not claim to have; funding-settlement times, for example, may affect order flow directly.

Controlling for 30 ten-second bars of lagged flow lowers $\hat{Q}$ further (Table S15, column $\hat{Q}_{30}$), consistent with persistent order splitting.

S13.6 Feedback across 30 perpetual markets

The companion paper estimates the same loop gain on one-minute bars of 30 Binance USD-M perpetuals spread across the liquidity ranking (2023–2024 training days). Measured feedback rises with depth: its Spearman correlation with log depth within 1% of the mid is +0.70 [0.41, 0.87], from about 0.012 in BTCUSDT and ETHUSDT to 0.0016 on average in the thinnest third, and the placebo pipeline gives $\hat{Q} \approx 0$ for every symbol. Feedback in this reduced-form sense is a feature of deep, algorithmically traded books.

Table S18: Zero fees on BTCUSDT: activity and measured feedback

	Zero fees introduced (2022-07-08)	Zero fees removed (2023-03-22)	Pooled
Log quote volume, $\hat{\beta}$	0.945 (0.074)	1.599 (0.080)	1.272 (0.084)
Observed-flow $\hat{Q}$, $\hat{\beta}$	0.00058 (0.00032)	−0.00091 (0.00060)	−0.00016 (0.00036)
p-value	0.078	0.140	0.650
Pair-days	320	320	640

$\hat{\beta}$ is the effect of zero fees on BTCUSDT relative to seven control pairs; date-clustered standard errors in parentheses. For the second event ZeroFee equals one before March 22, 2023.

S13.7 Two fee events

Binance set spot trading fees on BTC pairs to zero on July 8, 2022 and reinstated them on March 22, 2023. Each event changes the cost of trading BTCUSDT while leaving the seven other pairs unchanged, and the second reverses the first. For every pair and every day within 20 days of each event we compute daily quote volume and the observed-flow $\hat{Q}$, and estimate $Y_{it} = \alpha_i + \gamma_t + \beta\,\mathrm{ZeroFee}_{it} + e_{it}$ with pair and date fixed effects and date-clustered errors, separately for each event and pooled (Table S18).

The fee change moved activity enormously: zero fees multiplied BTCUSDT volume relative to the controls by $e^{0.945} \approx 2.6$ when they began, and their removal divided it by $e^{1.599} \approx 4.9$. Measured feedback did not follow. The two point estimates have opposite signs, neither is significant, and the pooled effect is -0.00016 with an interval of $[-0.00087, 0.00055]$, against a sample mean of 0.00037. A strong first stage with no second-stage response says that $\hat{Q}$ is not a by-product of trading activity, and the reversal design removes the main weakness of a single treated event. With one treated asset the interval remains wide, so this is evidence that $\hat{Q}$ does not scale with activity, not a proof of invariance.

S14 A finite-horizon decision that needs only the combined response

A separate note, `paper1_finite_horizon_supplement.pdf` in the public repository, works out one financial decision in full. An unlevered desk that warehouses inventory for H periods under a hard stress budget buys N reset experiments, pays for them out of the same capital, and sizes its position from an upper confidence bound on the summed price response over the horizon. With probability at least $1 - \gamma$ the position meets the stress budget for every bounded shock path; its profit shortfall is at most $AN^{-1/2} + Dc_HN$ for explicit constants A and D and cost c_H per experiment; and a stated inequality guarantees a positive expected

net value of the information. These guarantees hold uniformly over the positive cohort class, whatever the number and spacing of the cohorts. Under the same observations, telling apart two particular two-cohort economies whose rates are of order δ apart requires a number of experiments of order δ^{-6}, so at a fixed information price cohort identification becomes unaffordable while the decision keeps its positive value.

The lesson cuts both ways. Many risk decisions need only the combined response, which public data identify at the ordinary rate; class-level records are valuable only for decisions that depend on attribution, such as the targeted halt of Proposition 6.4. The note's proofs have not been independently reviewed, its numerical illustration is exploratory, and its priority is unconfirmed.

S15 Reproducibility

Every table, figure and number in this paper is mapped to the notebook or script and the result file that produce it in the reproducibility map, `ceiling_suite/papers/paper1/REPRODUCIBILITY_MAP.md` (also as `paper1_reproducibility_map.pdf`) in the public repository https://github.com/Pratikn03/Fianince, release tag `paper1-ssrn-v1`. The map also lists the replication notebooks in run order and the specification files of the pre-specified tests.